MILITARY HANDBOOK

INSTRUCTION FOR DESIGN, FABRICATION, AND

CONSTRUCTION/INSTALLATION OF SECURE ENCLOSURES



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ABSTRACT

This publication provides guidance for the design, fabrication, construction, and installation of secure enclosures and internal locking systems. The use of these low-cost structures will provide a secure, temporary, enclosure within an existing building for the storage of arms, ammunition, and explosives (AA&E). The Naval Civil Engineering Laboratory has developed this secure enclosure to provide adequate security for AA&E during nonworking hours in structures that otherwise do not meet the existing requirements of OPNAVINST 5530.13 for storage of AA&E.

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FOREWORD

This military handbook has been developed from an evaluation of facilities in the shore establishment, from surveys of the availability of new materials and construction methods, and from selection of the best design practices of the Naval Facilities Engineering Command (NAVFACENGCOM), other Government agencies, and the private sector. It uses to the maximum extent feasible, national professional society, association, and institute standards. Deviations from this criteria, in the planning, engineering, design, and construction of Naval shore facilities, cannot be made without prior approval of NAVFACENGCOMHQ Code 04.

Design cannot remain static any more than can the functions it serves or the technologies it uses. Accordingly, recommendations for improvement are encouraged and should be furnished to the Naval Civil Engineering Laboratory, Code L30, Port Hueneme, CA 93043, telephone (805) 982-5743.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

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PHYSICAL SECURITY CRITERIA MANUALS

Criteria Number	Title	PA
MIL-HDBK-1013/1	Physical Security of Fixed Land-based Facilities	NCEL
DM-13.02	Commercial Intrusion Detection Systems (IDS)	LANTDIV
MIL-HDBK-1013/3	Bolt-on Installation & Check-out Procedures for High-Security Hasp W/WO Anti-Intrusion Bar Cover	NCEL
MIL-HDBK-1013/4	Instruction for Design, Fabrication and Construction/Installation of Secure Enclosures	NCEL
MIL-HDBK-1013/5	Steel-Ply Wall Hardening Selection and Installation Guide	NCEL
MIL-HDBK-1013/6	High-Security Internal Locking System, Description, Operation, and Maintenance (PROPOSED)	NCEL
MIIL-HDBK-1013/7	Security Hardware Install., Operation & Maint. (PROPOSED)	NCEL

NOTE: Design manuals, when revised, will be converted to military handbooks.

This handbook is issued to provide immediate guidance to the user. However, it may or may not conform to format requirements of MIL-HDBK-1006/3 and will be corrected on the next update.

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Section 1: INTRODUCTION

1.1 Scope. This handbook provides sufficient information for the design, fabrication, and installation procedures for a secure enclosure within an existing building. Where adequate information for the system is provided in other Department of the Navy or commercial publications, such publication locations are referenced throughout this document. The information is not duplicated herein.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

1.2 Cancellation. This handbook supersedes no other publication.

1.3 General Safety Notices. Warnings and notes appearing throughout this publication are of paramount importance to personnel and system safety. The following general safety notices supplement the specific warnings appearing in this publication. They are recommended precautions that must be understood and applied during fabrication, installation, and maintenance of the system covered herein. Should a situation arise that is not covered in the general or specific safety precautions, the Commanding Officer or other authority will issue orders as deemed necessary to cover the situation.

1.3.1 Personnel Requirements. Do not fabricate or install alone. Under no circumstances shall the fabrication or installation of the secure enclosure be attempted without the immediate presence of an individual capable of rendering aid.

1.3.2 First Aid and Resuscitation. An injury, no matter how slight, should never go unattended. Always get first aid or medical attention immediately. Should an individual stop breathing, initiate resuscitation immediately. A delay could cause the loss of life.

1.3.3 General Precautions.

(a) All signs and markings that pertain to safety measures shall be displayed.

(b) Personnel shall always wear protective eyeshields, clothing, and safety shoes during fabrication, installation, and maintenance of the secure enclosure.

(c) Proper working stands shall be provided during fabrication of the secure enclosure panels.

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(d) All fabrication, installation, and maintenance actions on Naval shore facilities shall comply with appropriate policies, procedures, and guidelines provided in the Command Safety and Health Program, Naval Facilities Engineering Command (NAVFAC) Instruction 5100.11, as well as in all other pertinent NAVFAC 5100 series instructions.

1.3.4 Warnings. Warnings applying to the secure enclosure as covered in this publication are summarized below. These warnings are repeated throughout the publication following paragraph headings and preceding text where they apply.

WARNINGS

A respirator should be worn when welding galvanized steel parts indoors. Fume (zinc oxide) may cause gastric distress. (Page 23)

Grating used in fabricating panels and finished panels is heavy, 221 to 315 pounds. Care shall be taken in handling the grating and panels to ensure they are not dropped. (Page 23)

Panels must be securely braced in the upright position before bolting. Injury to personnel and damage to panels may occur if a panel falls. (Page 37)

Avoid prolonged contact with, or inhalation of, cleaning solvents. Avoid use near heat or open flame, and provide adequate ventilation. (Page 45)

1.4 Related Technical Documents. The following drawings of various secure enclosure components are required for use with this document.

1.4.1 Drawings. Specific drawings include the following:

(a) Physical Security: Riveted Steel Grating Cages - Configurations and Assembly. NAVFAC drawing number 6155276.

(b) Physical Security: Riveted Steel Grating Cages - Panels, Supports, and Surface Bolts. NAVFAC drawing number 6155277.

(c) Physical Security: Welded Steel Grating Cages - Configurations and Assemblies. NAVFAC drawing number 6155278.

(d) Physical Security: Welded Steel Grating Cages - Panels, Supports, and Surface Bolts. NAVFAC drawing number 6155279.

(e) Physical Security: Riveted Steel and Welded Steel Grating Cages - Internal Locking System. NAVFAC drawing number 6155318.

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(f) Physical Security: Riveted Steel and Welded Steel Grating Cages - Internal Locking System, Face Plates. NAVFAC drawing number 6155319.

These drawings are available from:

Naval Civil Engineering Laboratory
Physical Security Division (Code L56)
Port Hueneme, California 93043-5003

A Naval Ammunition Production Engineering Center (NAPEC) drawing is also recommended:

(g) High-Security Hasp, Hinged, or Sliding Horizontal, RH MK2 MOD 8. NAPEC drawing number 0957 (three sheets).

This drawing is available from:

Naval Weapons Support Center
Code 208
Crane, IN 47522-5020
AV 482-1879

1.4.2 Military Specifications. Certain military specifications are recommended.

(a) MIL-G-81322: Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code Number G-395.

(b) MIL-M-7866B: Molybdenum Disulfide Technical Lubrication Guide.

Department of Defense activities may obtain copies from:

Commanding Officer
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Section 2: SECURE ENCLOSURE SYSTEM DESCRIPTION

2.1 Safety Precautions.

2.1.1 Warnings. Warnings appearing throughout this publication are of paramount importance to personnel safety. Prior to any attempt to fabricate, install, maintain, troubleshoot, or repair any part of the secure enclosure (Figure 1), all warnings shall be thoroughly reviewed and understood. Refer to Section 1: Introduction of this publication for a complete listing of warnings. The following paragraphs define warnings and notes as they are used in this publication.

WARNING

Identifies an installation or maintenance procedure, practice, condition, statement, etc., which, if not strictly followed, could result in death or serious injury to personnel.

NOTE

Highlights certain fabrication, installation, or maintenance statements that are essential but not of known hazardous nature as indicated by warning.

2.1.2 Personnel Responsibilities. Personnel responsible for the secure enclosures shall become thoroughly familiar with and frequently review the following specific safety precautions. These precautions apply to both personnel and equipment and are considered supplemental to the general precautions and specific warnings listed in Section 1: Introduction.

(a) Do not fabricate, install, or maintain the secure enclosure without first reading and understanding the appropriate portions of this publication.

(b) Personnel shall always wear protective eyeshields, clothing, and safety shoes when fabricating, installing, or maintaining the secure enclosure.

2.2 Secure Enclosure System Background. With the increased awareness of the threat of theft to conventional arms, ammunition, and explosives (AA&E), the requirement for a low-cost, high-security enclosure for temporary AA&E storage has been recognized. The secure enclosure described in this manual satisfies this requirement. The secure enclosure (Figure 1) can be constructed in a variety of configurations and sizes. When no longer required for temporary storage, the enclosure can be disassembled and stored for future use.

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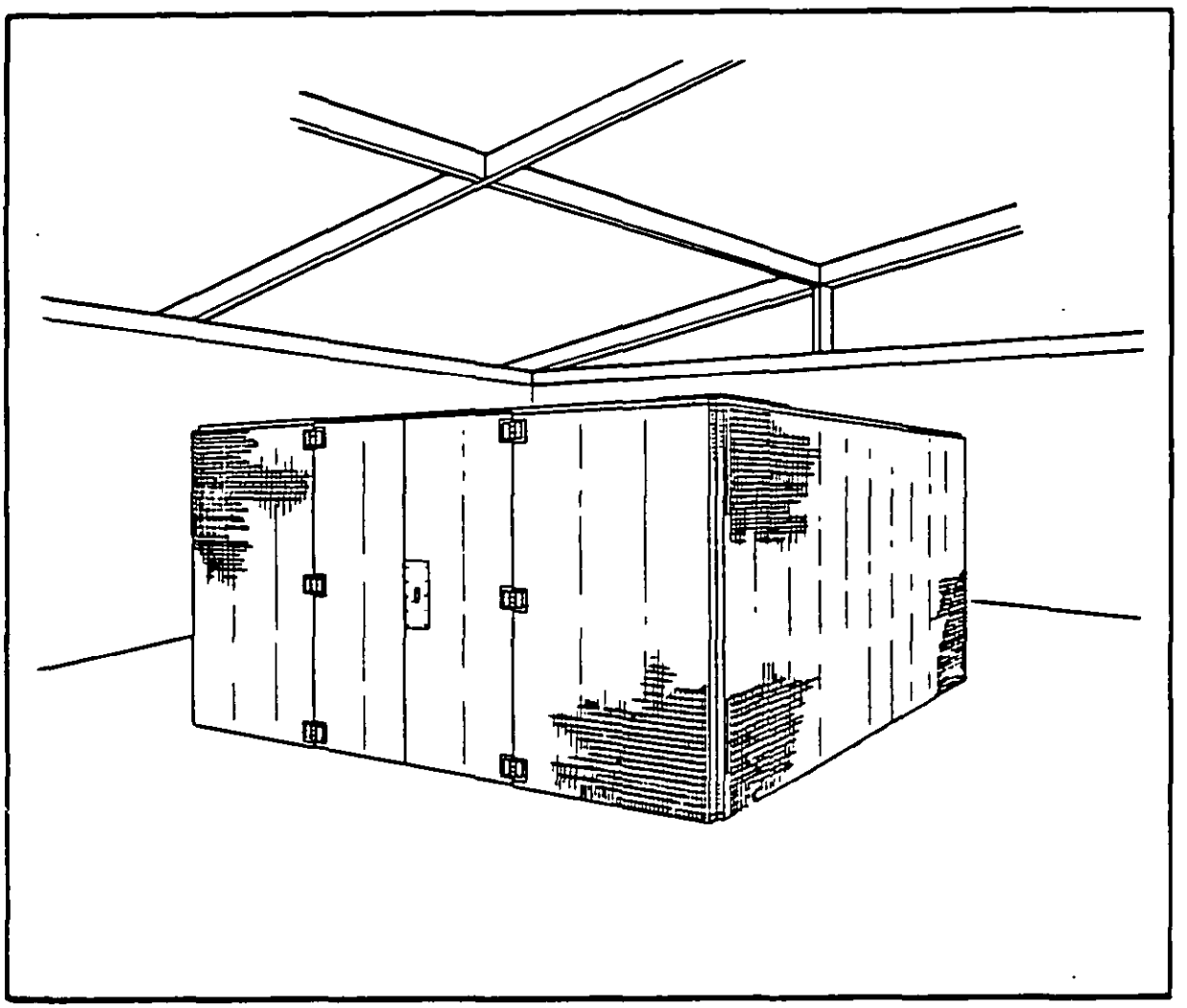


Figure 1. Secure Enclosure

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2.3 System Description.

2.3.1 General. The secure enclosure is a low-cost system of 10- by 2-foot riveted or welded steel panels (Figures 2 and 3) bolted together to form any size enclosure in 2-foot increments, up to 20 feet on the shorter side(s). There is no limit to the length of the enclosure. However, a spanner beam must be used with each 10-foot increment to support the roof panels. Doors assembled from similar panels may be single or double leaf. The enclosures are easily installed and may be erected in existing buildings that otherwise would not be adequate for secure storage. Use of the secure enclosure will eliminate the requirement for returning AA&E to approved secure storage during nonworking hours as required by OPNAVINST 5530.13, Physical Security for Sensitive Conventional AA&E. The secure enclosure (when bolted to the floor or wall of the structure), and the enclosure door(s), (secured by the required locking system), provide penetration resistance consistent with high-security locking systems.

There are two approved locking systems that can be used to secure the enclosure. They are: the internal (Figure 4) and external (padlock) (Figure 5) locking systems. The two systems are described in Paragraphs 2.3.4 and 2.3.5. The internal locking system offers the advantage of having the locking system located inside the secure enclosure, thereby decreasing the locking system's vulnerability to forced attack. The internal system is simpler to operate because the key is inserted in the horizontal position rather than vertical, as required in unlocking the padlock. Additionally, the lock does not require removal each time the door is opened. The internal system should require less maintenance since no handling of the lock is required. The external padlock requires removal and reinstallation each time the enclosure is opened and closed.

2.3.2 Riveted Grating Panels. The riveted grating panels (Figure 2) consist of commercially available riveted steel grating that is welded to 2-inch-square perforated steel tubing to form 10- by 2-foot panels. The panels are joined by bolts to form the walls, roof, and doors of the enclosure. The riveted panels are more resistant to penetration than the welded steel grating; however, they are considerably heavier. Each panel weighs 315 pounds.

2.3.3 Welded Grating Panels. The welded grating panels (Figure 2) consist of commercially available welded steel grating that is welded to 2-inch-square perforated steel tubing to form 10- by 2-foot panels. The panels are joined in the same method as the riveted panels described in Paragraph 2.3.2. Each panel weighs 221 pounds.

2.3.4 Internal Locking System. The internal locking system is installed in a housing on the inside of the secure enclosure door. The major components of the system are the keylock boltwork actuating mechanism and the boltworks. The only parts of the system visible on the outside of the enclosure are the face plate, keyway for the lock, and the actuating handle for the boltworks (Figure 4). The system is operated by placing the key in the door keyway and turning counterclockwise. This action unlocks (releases) the boltwork mechanism, allowing its operation. Turning the actuating handle clockwise operates

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the boltworks, which disengages a locking bar, allowing the door to be opened. As described in Paragraph 2.3.1, the internal locking system is less vulnerable than the external locking system.

2.3.5 External Locking System. The external locking system is a simple high-security locking system consisting of the NAPEC 0957/0958 High-Security Hasp (Figure 5) and padlock in accordance with MIL-P-29151, Lock and Lock Sets, Exterior, Ordnance, High Security; and MIL-P-43607, Padlock, Key Operated, High-Security Shrouded Shackle. The hasps are welded on the outside of the enclosure and secured by the padlock.

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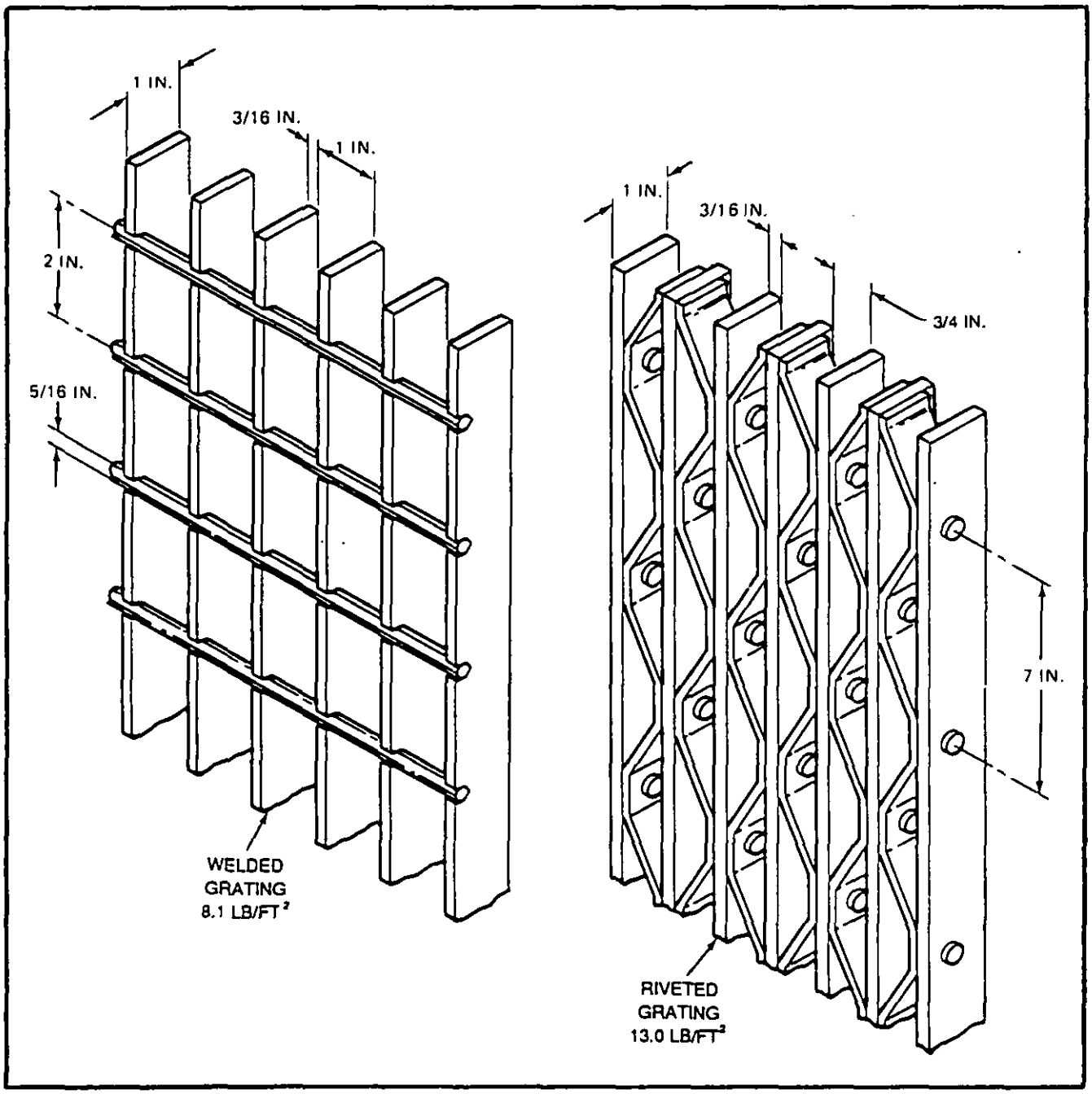


Figure 2. Typical Gratings

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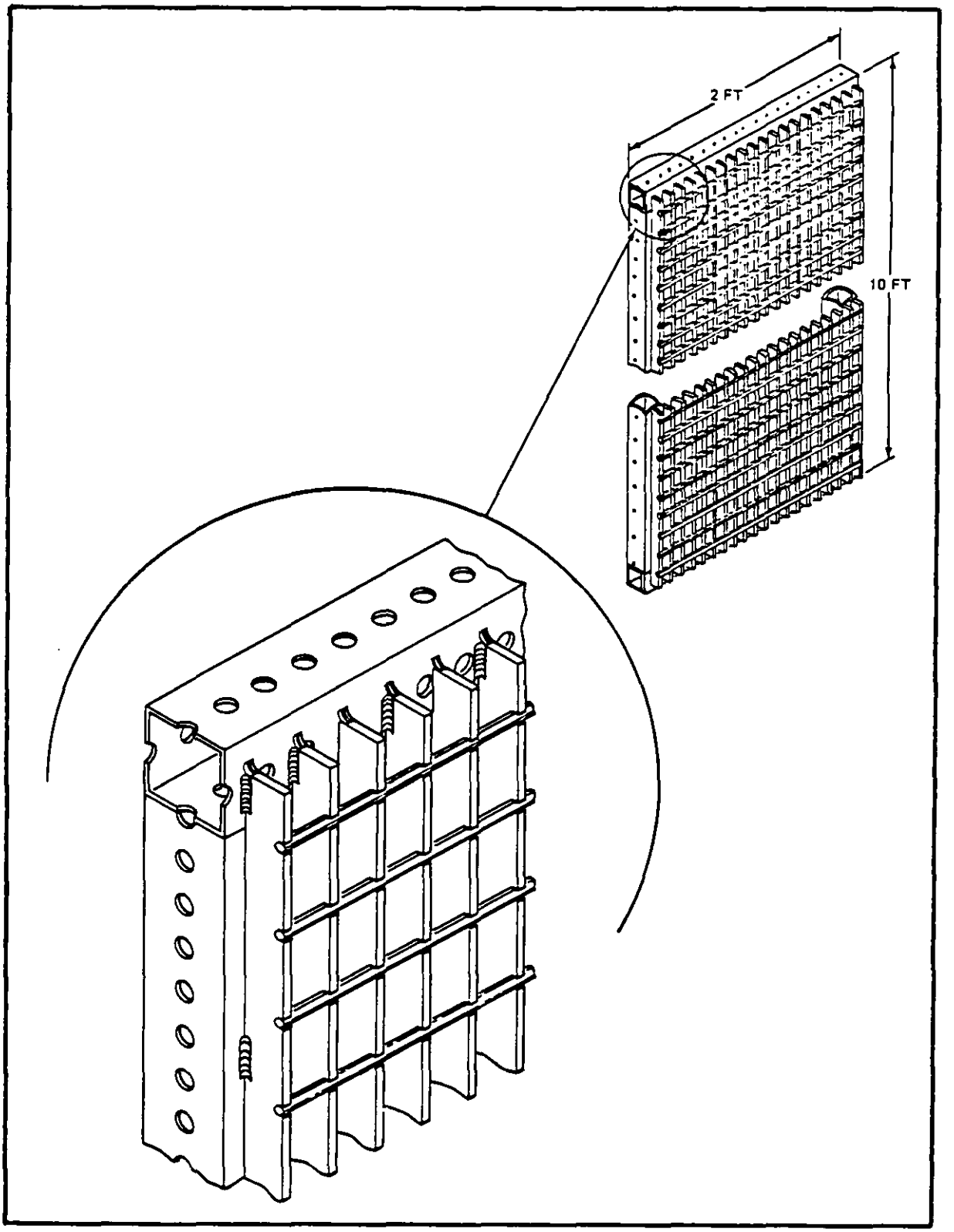


Figure 3. Typical Standard Panel

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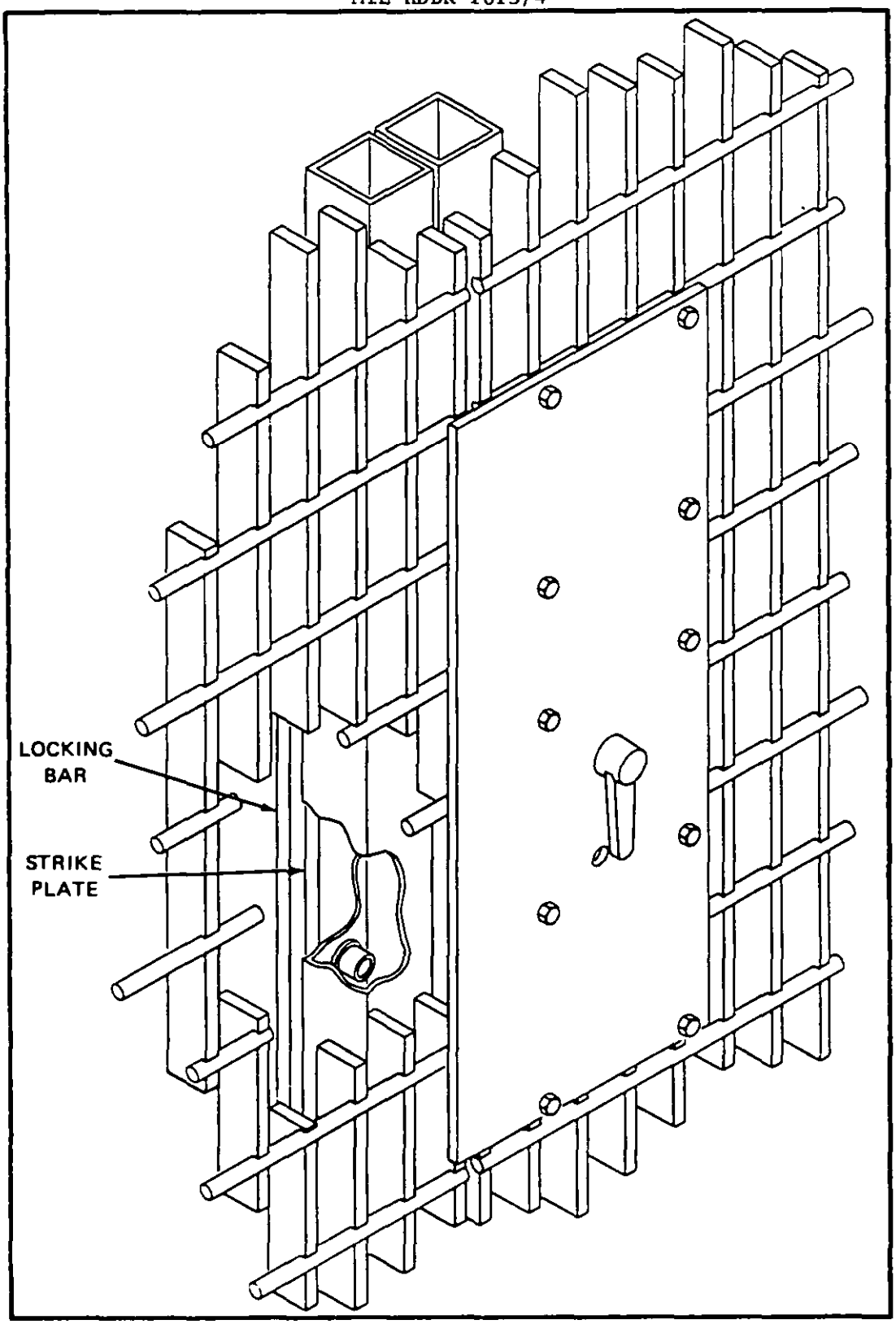


Figure 4. Internal Locking System

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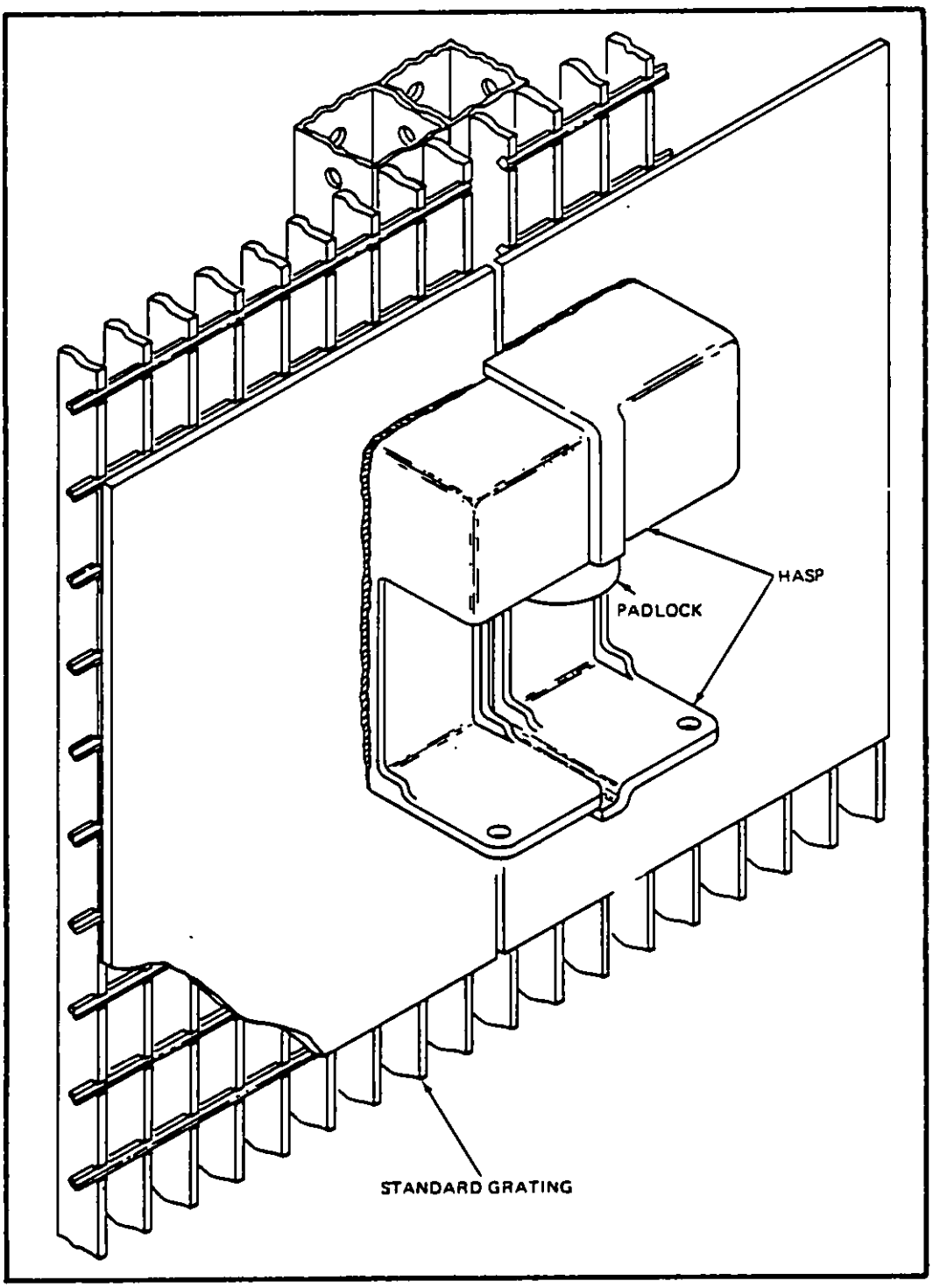


Figure 5. External Locking System

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Section 3: DESIGN

3.1 Design Selection. As described in Section 2, the secure enclosure can be assembled in a variety of configurations up to 20 feet wide and of any length (in 2-foot increments). The enclosures may use existing walls, provided the walls meet or exceed the minimum requirements of OPNAVINST 5530.13. Floors must have a penetration resistance equal to or greater than the type of grating selected. A five-sided enclosure (Figure 1), consisting of four walls and a roof, will be required when existing walls do not meet minimum requirements. In addition to the five-sided enclosure, other basic configurations are as follows:

(a) Four-sided, three sides and a top, against an existing wall (Figure 6).

(b) Three-sided, two sides and a top located in a corner of a building (Figure 7).

(c) Two-sided, one side and a top located in the end of a narrow room (Figure 8).

(d) Stand-off, to provide protection to low penetration-resistant doors or cave entrances (Figure 9).

3.2 Structural Loading Considerations.

3.2.1 General. After design of a secure enclosure configuration is complete and a location for installation is selected, the static and dynamic floor loads must be taken into consideration. Static loading is defined as the weight of the enclosure on the existing building floor. Dynamic loads are horizontal forces brought on a structure by seismic activity. Seismic activity will affect secure enclosure designs that need to be bolted to an existing building wall.

3.2.2 Weight Calculations. The weight of a secure enclosure is dependent on the type of grating panels and enclosure design size. The number of panels required varies with the design, and the number of spanner beams required varies with the geometry of the enclosure. The range of enclosures sizes and corresponding maximum floor loading per square inch is contained in Table 1. The weight of the enclosure is calculated and used to check the load capacity of the existing floor. The enclosure must be situated on a concrete floor on grade, or on floors capable of supporting forklift loads. Wood floors or an unreinforced, suspended slab generally will not support the weight of the smallest size secure enclosure unless the floor is strengthened. Strengthening a floor to support the required loading may defeat the overall purpose of installing a secure enclosure. Therefore, the secure enclosure should be installed only on floors that do not require modification. Punching shear, caused by a concentrated load that exceeds the shear strength of a concrete

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Table 1
Floor Load Range for Enclosures in Pounds Per Square Inch

			Maximum Floor Load (psi)		
Enclosure Arrangements (feet)	Number of Walls	Number of Spanner Beams	Riveted Panels (315 pounds/ panel)	Welded Panels (221 pounds/ panel)	Location of Maximum Load
10 x 10	4	0	7	5	walls
10 x 10 w/roof	4	0	10	7	walls
20 x 20	4	0	7	5	walls
20 x 20 w/roof	4	1	215	157	tubing column**
20 x 20	3	0	7	5	walls
20 x 20 w/roof	3	1	215	157	tubing column**
60 x 20	4	0	7*	5*	walls
60 x 20 w/roof	4	5	215*	157*	tubing column**

*These values apply to all larger enclosures.
**Spanner beams supporting roof panels transmit roof loading to the two 2- by 2-inch tubing columns that support the spanner beams.

slab, should be analyzed in locations where large loads are supported on small areas. For example, the two 2- by 2-inch columns supporting a roof spanner beam carry the partial weight of the roof and concentrate that weight on a small portion of the floor directly under the columns. Maximum floor loading must also consider the weight of the materials to be stored in the enclosure and the associated handling equipment. Figure 10, developed from the 1982 edition of the American Institute of Architects, is provided to determine minimum floor (slab) thickness requirements. The thickness requirements can be estimated by entering the graph from the left at the maximum weight to be stored (or wheel footprint of material handling equipment) and entering the bottom of the graph at the minimum floor contact area of the materials (or footprints). Floor thickness calculations must be checked by a structural engineer. A 4-inch-thick slab on grade will support any secure enclosure configuration contained in this publication.

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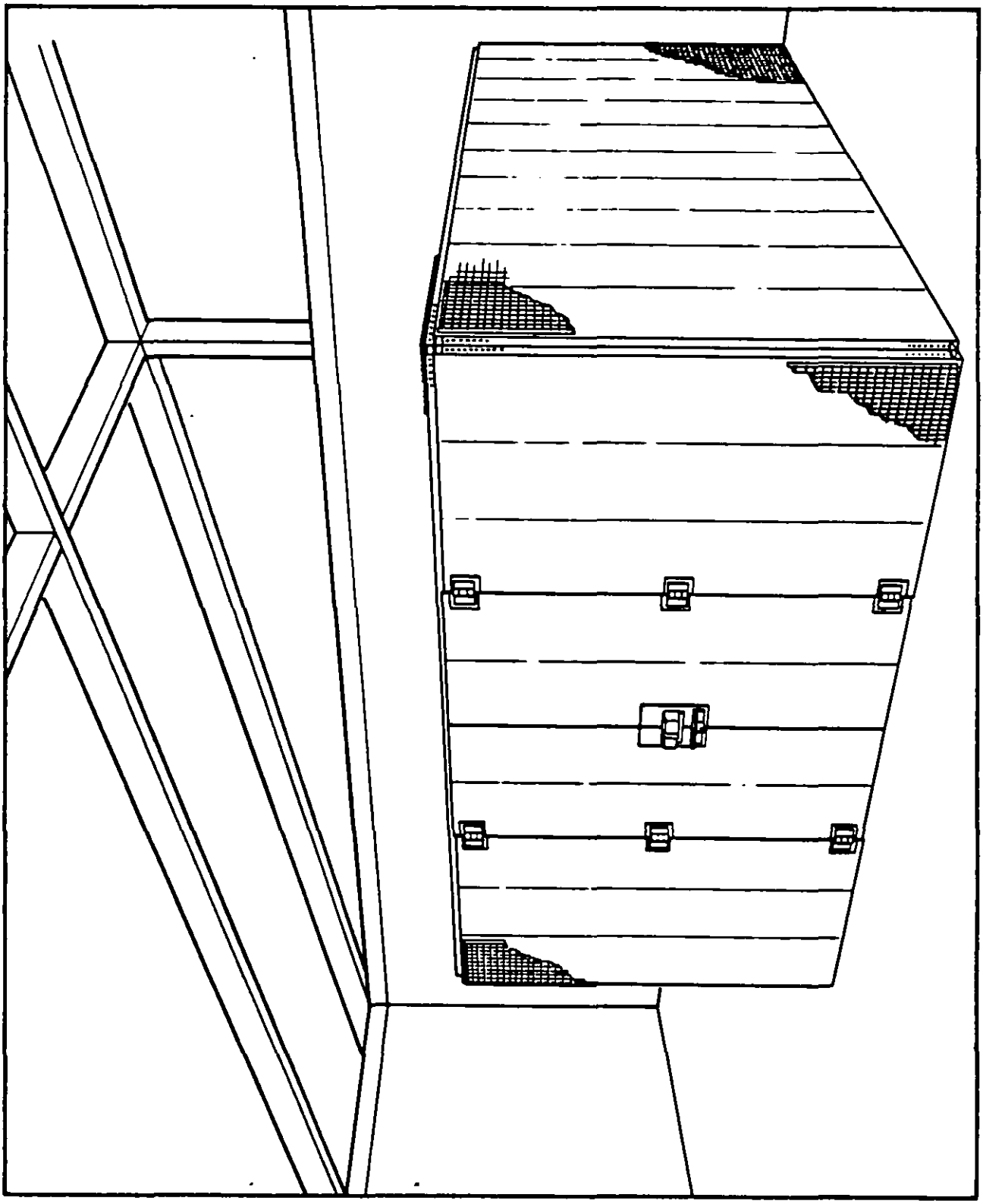


Figure 6. Four-Sided Enclosure

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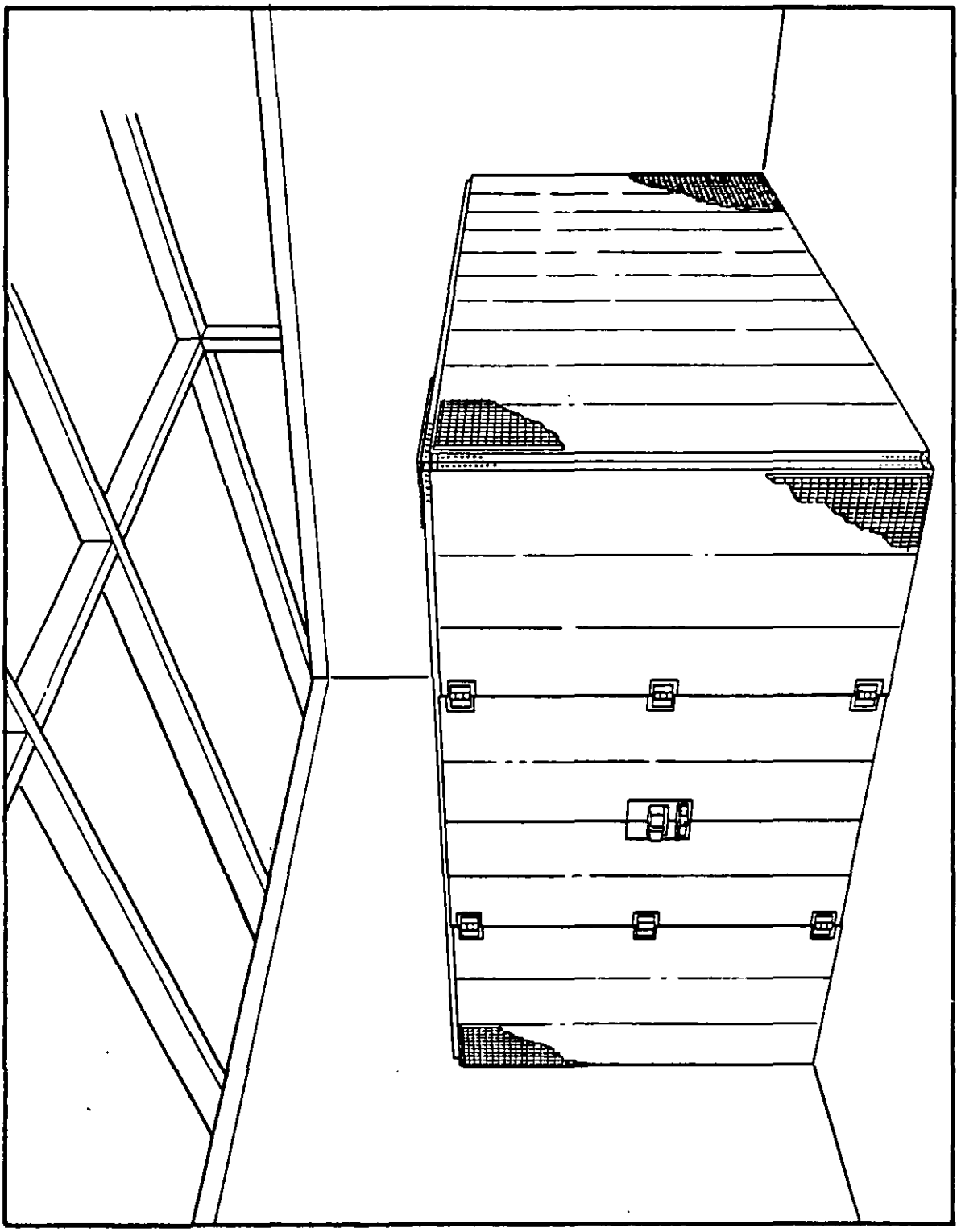


Figure 7. Three-Sided Enclosure

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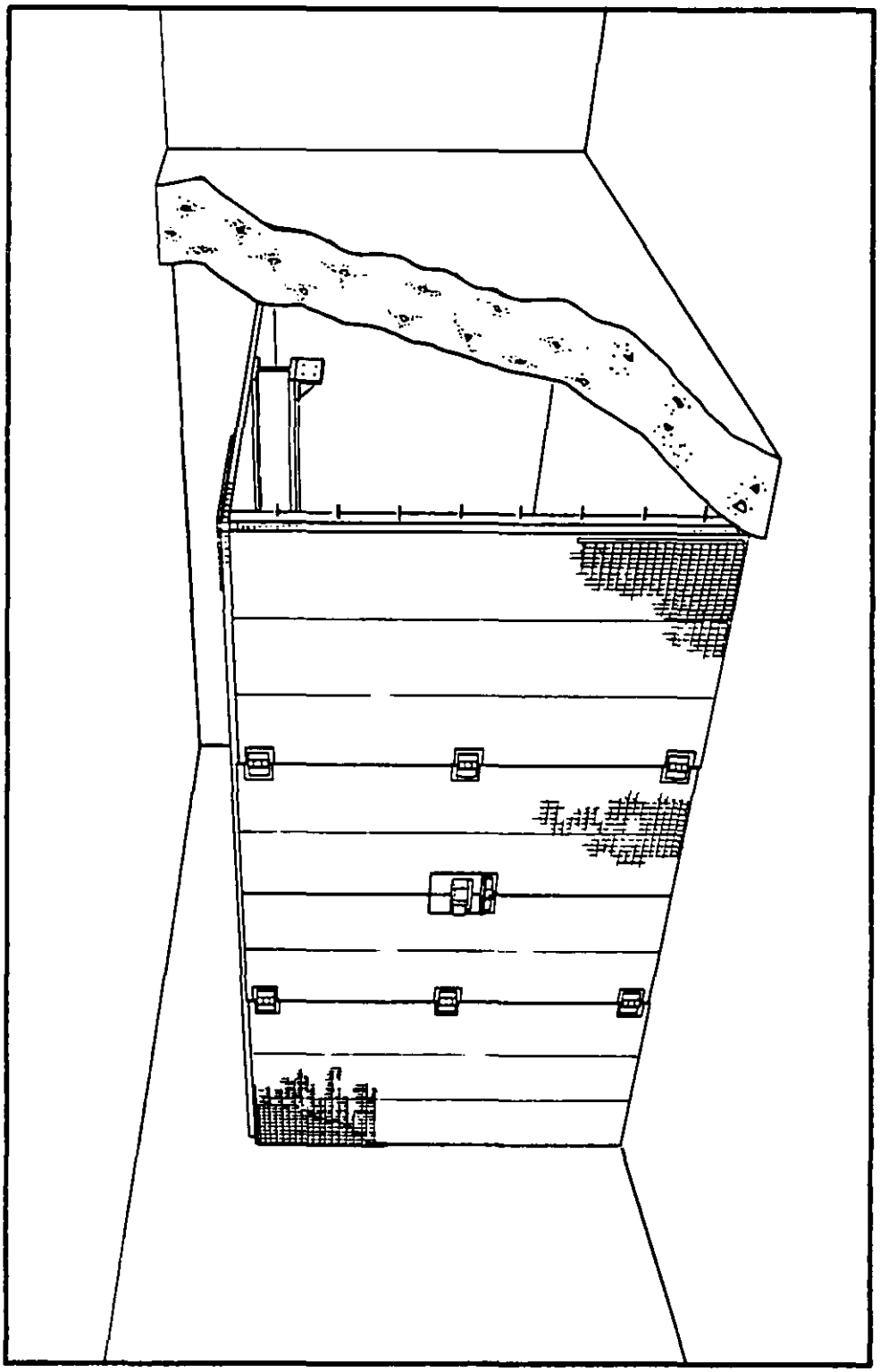


Figure 8. Two-Sided Enclosure

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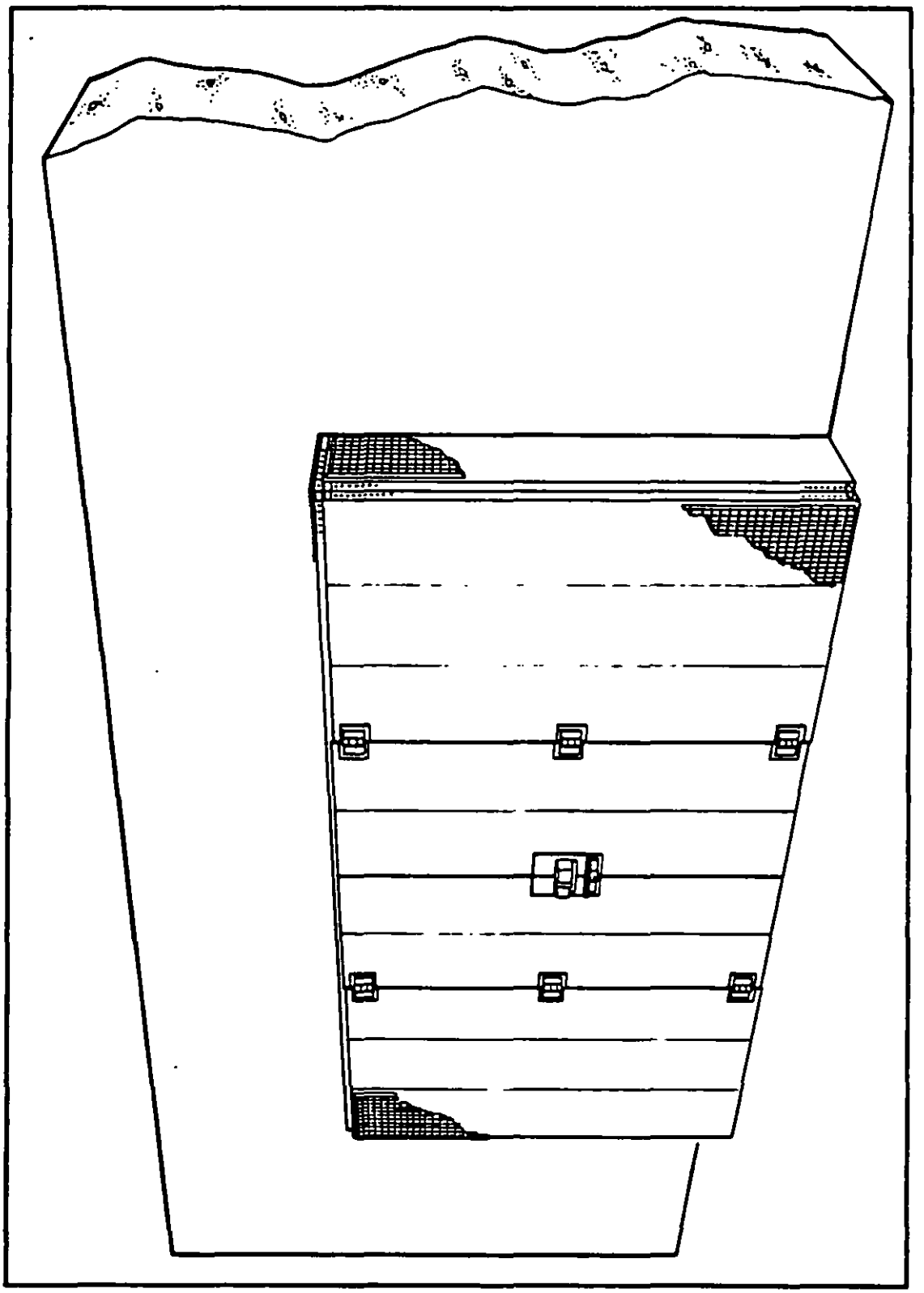


Figure 9. Stand-Off Enclosure

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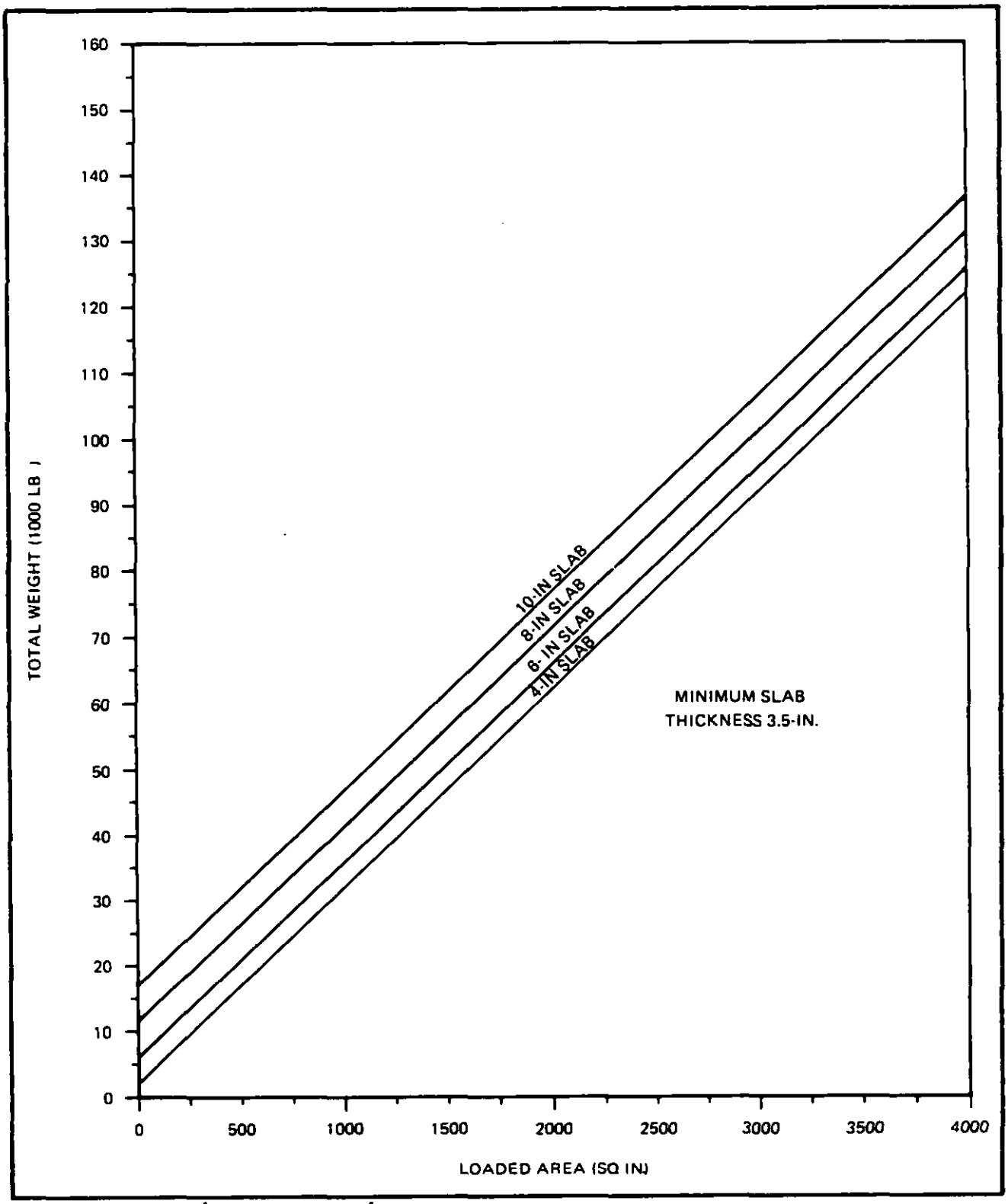


Figure 10. Maximum Loads for Slabs (Flexural Tensile Stress = 300 psi)

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3.2.3 Dynamic Loads. These loads are horizontal forces that act on secure enclosures attached to existing building wall(s). In general, secure enclosures will be attached to reinforced concrete walls or concrete block not less than 8 inches thick as required by OPNAVINST 5530.13. Concrete expansion anchors are used to attach the enclosure to an existing wall. Bolts of 3/8-inch diameter are used on 16-inch centers along the secure enclosure roof and wall panels. This attachment to the existing building will cause the enclosure to move with the building during seismic disturbances. Earthquake zones should be assessed using Figure 11. The type of stress on bolts caused by seismic disturbances depends on the enclosure configuration and direction of motion caused by the disturbance. For example, if an enclosure is fastened to the corner walls of a concrete structure, and the motion caused by an earthquake is parallel with one of the walls, the stress on the bolts in that wall will be in shear. Bolts in the wall perpendicular to the motion will be in tension. The shear and tension on the bolts should be checked and compared with manufacturer's specifications or Table 2. Information on seismic design for specific sites is available in the Technical Manual for Seismic Design for Buildings (Navy NAVFAC-P355, Army TM-5-809-10, Air Force AFM 88-3, Chap 13). Structural integrity of building walls should be analyzed for possible complications in shear and tension at the fasteners. The analysis should be checked by a structural engineer.

3.3 Design Procedure. Before a secure enclosure is designed, it must first be determined whether a suitable building is available to hold the volume and weight of the material required for storage. Additionally, the availability of security forces as required by OPNAVINST 5530.13 must be determined. If all requirements are satisfied, the following procedural steps should be followed in designing the secure enclosure for installation in an existing structure:

Table 2
Allowable Shear and Tension on Bolts

		Minimum Concrete Strength (in psi)*		
Bolt Diameter (in.)	Minimum Embedment (in.)	Shear		Tension
		2,000	3,000	2,000 to 5,000
1/4	2-1/2	500	500	200
3/8	3	1,100	1,100	500
1/2	4	2,000	2,000	950
5/8	4	2,750	3,000	1,500
3/4	5	2,940	3,560	2,250
7/8	6	3,580	4,150	3,200
1	7	3,580	4,150	3,200
1-1/8	8	3,580	4,500	3,200
1-1/4	9	3,580	5,300	3,200

*Values are for natural stone aggregate concrete and bolts of at least A307 quality.

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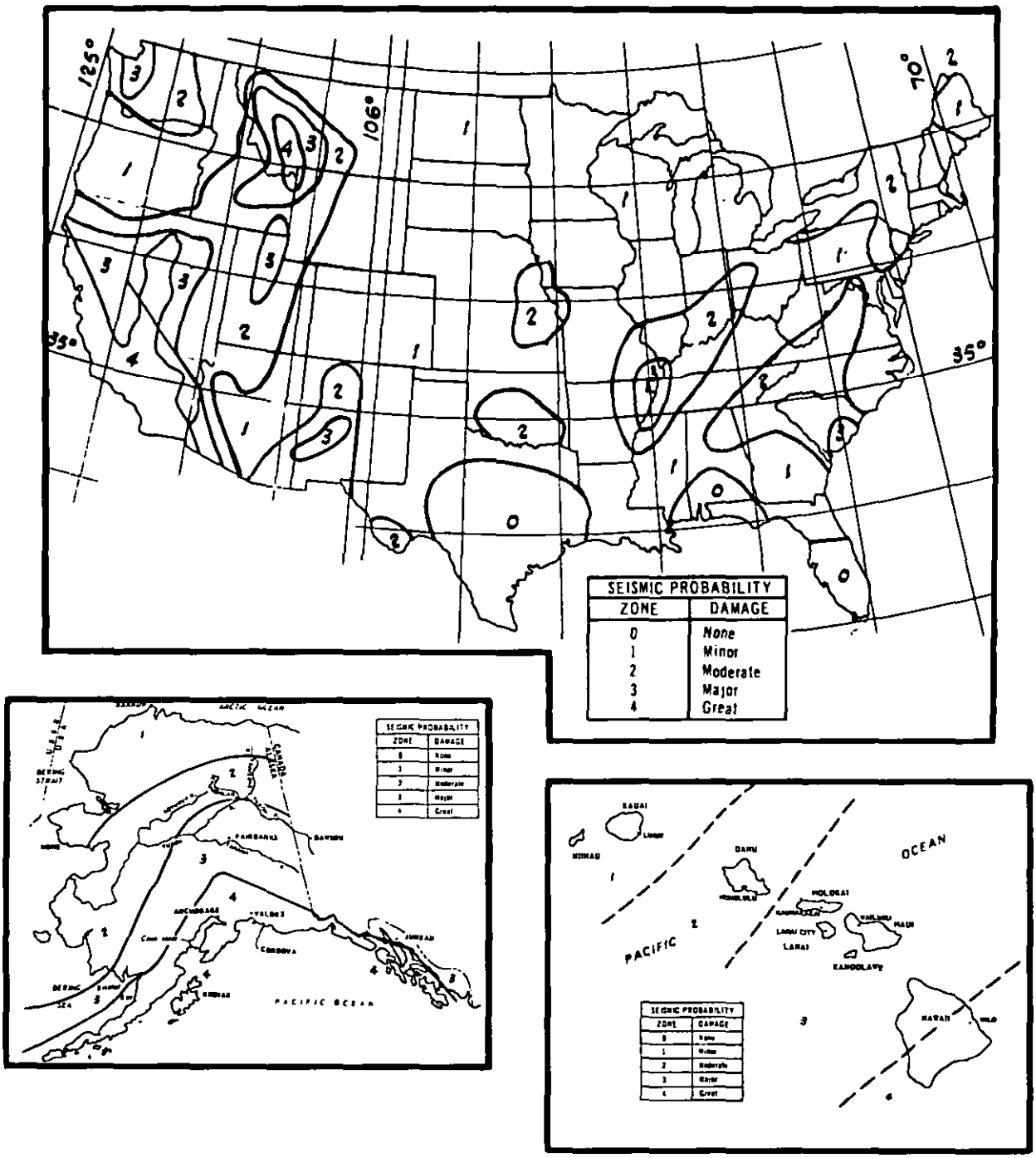


Figure 11. Seismic Zone Map of the United States

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(a) Determine size of enclosure required for volume of material to be stored.

(b) Check to determine suitability of existing structure walls to serve as wall(s) of enclosure.

(c) For structures with walls unsuitable for use as an enclosure wall, use five-sided, free-standing enclosure (Figure 1) installed several feet away from walls.

(d) For structures with walls adequate for use as enclosure walls, select a structure site that can make maximum use of existing structure walls. Refer to Figures 6 through 9 for basic configurations.

(e) Determine door location and type of door (one- or two-leaf) to satisfy material handling requirements.

(f) Using Table 3, determine type of grating required for arms, ammunition, and explosives (AA&E) risk category. Table 4 contains test results from penetration attempts on two available gratings.

(g) Determine type of locking system to be used on enclosure, taking into consideration increase in security gained by installation of the internal locking system.

Table 3
Security Risk Categories*

Category	AA&E (in part)	Type of Grating Required
I	Hand-held missiles, rockets, and their ammunition	Riveted steel**
II	Light automatic weapons, grenades, and mines	Riveted or welded steel
III	Launch and mortar tubes, flame throwers, 20-mm and larger ammunition	Riveted or welded steel
IV	Rifles, hand guns, ammunition without explosive projectile	Riveted or welded steel

*For complete definition of AA&E risk category, see OPNAVINST 5530.13, Physical Security Instruction for Sensitive Conventional AA&E.
**Storage of risk category I will be approved by NAVSEA on a case-by-case basis.

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Table 4
Multilevel Attack Tests

Grating Sample	Attack Tools	Actual Working Time (min)	Total Working Time (min)	Remarks
Riveted steel	Handtool*	5.0	Unknown	Stopped**
	Jaws of Life	6.5	15.0	Opening made
	12-inch abrasive saw	6.5	6.75	Opening made
	Oxy-acetylene torch	2.5	3.08	Opening made
	Burn bar	1.8	3.7	Opening made
Welded steel	Handtool*	10.0	39.0	Opening made
	Jaws of Life and bolt cutter	5.5	9.0	Opening made
	12-inch abrasive saw	1.3	1.8	Opening made
	Burn bar	1.0	1.0	Opening made

*Combination of hand attack tools.
**Attack stopped because tools were ineffective.

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Section 4: FABRICATION

4.1 Introduction.

4.1.1 General. This section describes how each component of the secure enclosure is fabricated. The tools, equipment, and materials required to fabricate each component are identified. All components requiring fabrication in accordance with the drawings mentioned must be fabricated in a shop properly equipped for metal fabricating and machining.

4.1.2 Materials and Tolerances. All components of the enclosure are to be fabricated using only the materials prescribed in this publication. Substitution of other materials could cause a security vulnerability. The tolerances prescribed on the drawings must be closely adhered to during fabrication. If tolerances are not followed, the final fitting and assembly of the component parts will be adversely affected.

4.2 Panels.

WARNING

A respirator should be worn when welding galvanized steel parts indoors. Fumes (zinc oxide) may cause gastric distress.

WARNING

Grating used in fabricating panels and furnished panels is heavy, 221 to 325 pounds. Care shall be taken in handling the grating and panels to ensure they are not dropped.

4.2.1 Wall and Roof Panel Fabrication. The wall and roof panels are identical and are fabricated with either the riveted or welded steel grating in accordance with Naval Facilities Engineering Command (NAVFAC) drawing 6155277 or 6155279. Equipment and tools required for fabrication are contained in Table 5. Material (hardware) requirements are listed on the bill of materials on the drawings and are summarized in Table 6. Panel frames are fabricated from 2- by 2-inch, 12-gauge, perforated, Unistrut steel tubing. The tubing is cut and welded in accordance with drawings to form a 2- by 10-foot frame. The galvanized commercially available riveted or welded steel grating is then welded to the frame as specified.

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Table 5
Equipment Requirements for Panel Fabrication

Quantity	Description
1	140-amp AC/4,000-watt welder/power plant
4	6-inch C-clamps, malleable iron
1	5-1/2- x 5-1/2-inch power hacksaw
1	File, half round bastard, 12 inch
1	Sandblasting services
1	Paint sprayer with air compressor

Table 6
Hardware Requirements

Item	Application	Size (ft)	Quantity	Source
Riveted grating	Risk category I*, II, III, and IV	2 x 10	As required	Commercial (see drawing 6155277)
Welded grating	Risk categories II, III, and IV	2 x 10	As required	Commercial (see drawing 6155279)
Perforated tubing	All panels	---	As required	Commercial (see applicable drawing)

*Secure Enclosures for risk Category I will be approved by NAVSEA on a case-by-case basis.

4.2.2 Door Panel Fabrication. Fabrication methods for door panels are the same as for wall and roof panels. The difference between the panels, as shown in the drawings, is that door panels are 1/16-inch narrower than the other panels. This allows the door to swing open on its hinges. If two or more panels are used to form a door leaf, the panels are bolted together using the splice bar shown in NAVFAC drawing 6155276. Weld three hinge mounting plates (3-3/4 by 1/4 by 6-1/2 inches) to the grating before welding the grating to the tube frame.

4.2.2.1 Locking System Requirements. If the external locking system is used, the 10-1/2- by 1/4- by 16-inch hasp mounting plate can be welded to the grating before or after welding the grating to the tube frame. Selecting

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the internal locking system will require fabrication and installation of the lock housing, face plate, and commercially available keylock and boltwork mechanism in accordance with NAVFAC drawings 6155318 and 6155319. The face plate and lock housing are attached to the panel by bolts and welding.

4.2.2.2 Double-Door Surface Bolt Fabrication. When a double-door configuration is used, the inactive door panel will have two door surface bolts installed as indicated in NAVFAC drawings 6155276 and 6155278. The hand-retractable plunger (commercially available) is welded to the fabricated housing as indicted in the drawing. One unit is bolted to the inside top edge and the other to the inside bottom edge of the inactive leaf (Figure 12).

4.3 Locking Systems.

4.3.1 Internal Locking System. All major components of the internal locking system, with the exception of the keylock and boltwork actuating mechanism, require shop fabrication. Equipment requirements for fabricating the internal locking system are contained in Table 7. The keylock and boltwork mechanism are available as a single unit from LAGARD Inc., Torrance, California. The major components of the internal locking system that require fabrication are the face plate, housing, anti-drill plate, guide block, actuating bar, locking bar, strike plate, and shroud (Figure 13). The Bill of Materials and fabricating details are shown in NAVFAC drawings 6155318 and 6155319. Components of the system are illustrated in Figures 14 through 18. The face plate and housing should be the first components to be fabricated. Those components, along with the furnished keylock and boltwork actuating mechanism, may be installed on the door panel during door panel fabrication. (All other components are to be assembled during installation of the secure enclosure.) The entire internal locking system may be installed (retrofitted) on an assembled enclosure.

4.3.2 External Locking System. If the external locking system is used, a hasp mounting plate will have to be welded to the door panels during panel fabrication as indicated in Paragraph 4.2.2. The hasp will be welded to the mounting plate when the door panels are installed on the enclosure.

4.4 Assembly Hardware Fabrication. Several hardware items require in-shop fabrication. Details for fabrication are contained in NAVFAC drawings 6155277 and 6155279. The required items of hardware and quantities required per panel are listed as follows (total quantities required for assembly of an enclosure will vary depending on the enclosure design):

(a) Clip Angle: A 2-inch support angle cut from a 3- by 2- by 1/4-inch angle. Two clip angles are required per panel for panel to floor attachment.

(b) Corner Angle: A 9-foot by 11-3/4-inch angle cut from a 2- by 2- by 5/16-inch angle. One corner angle is required for each corner, either panel-to-panel wall corner or panel to corner.

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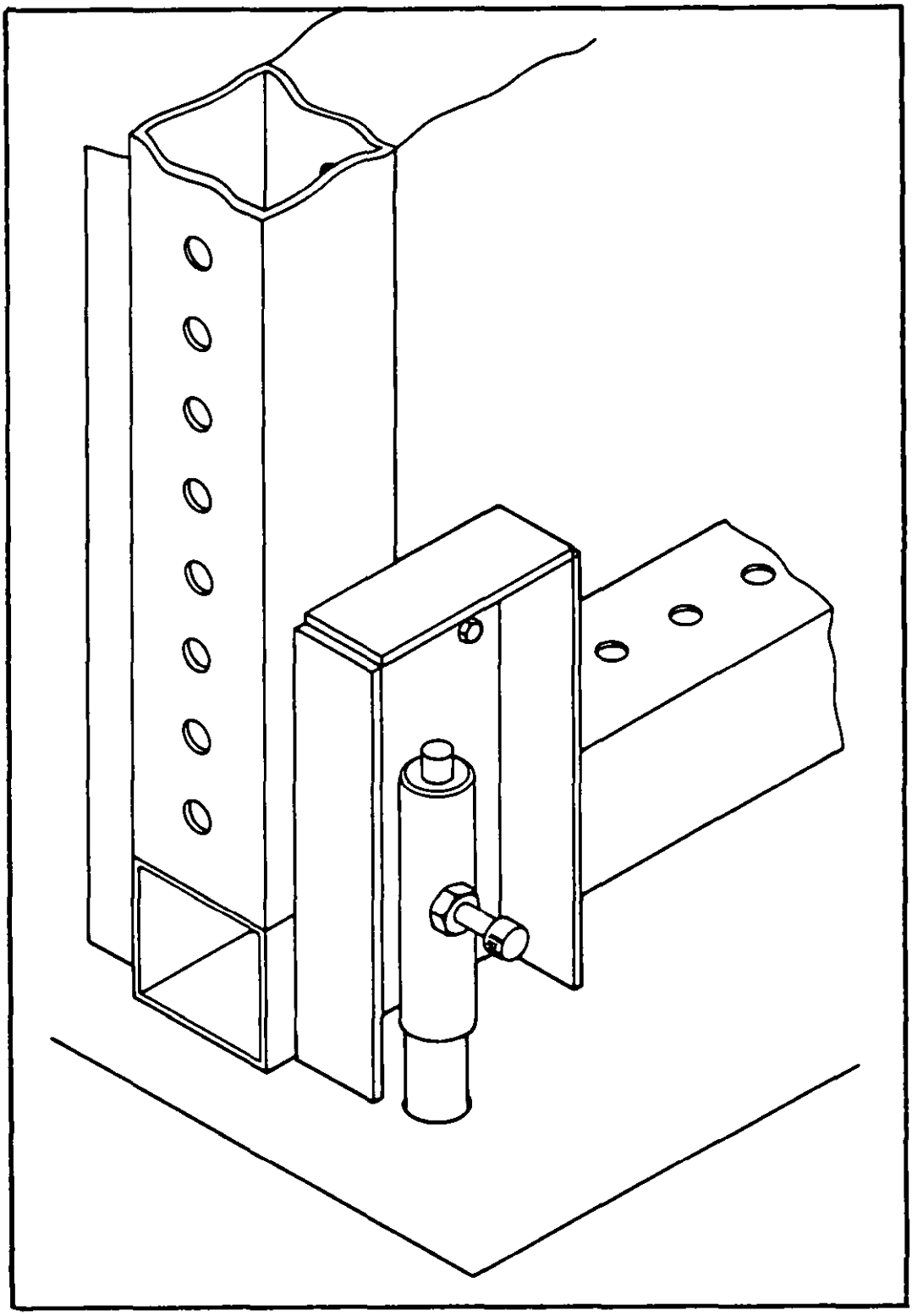


Figure 12. Door Surface Bolt

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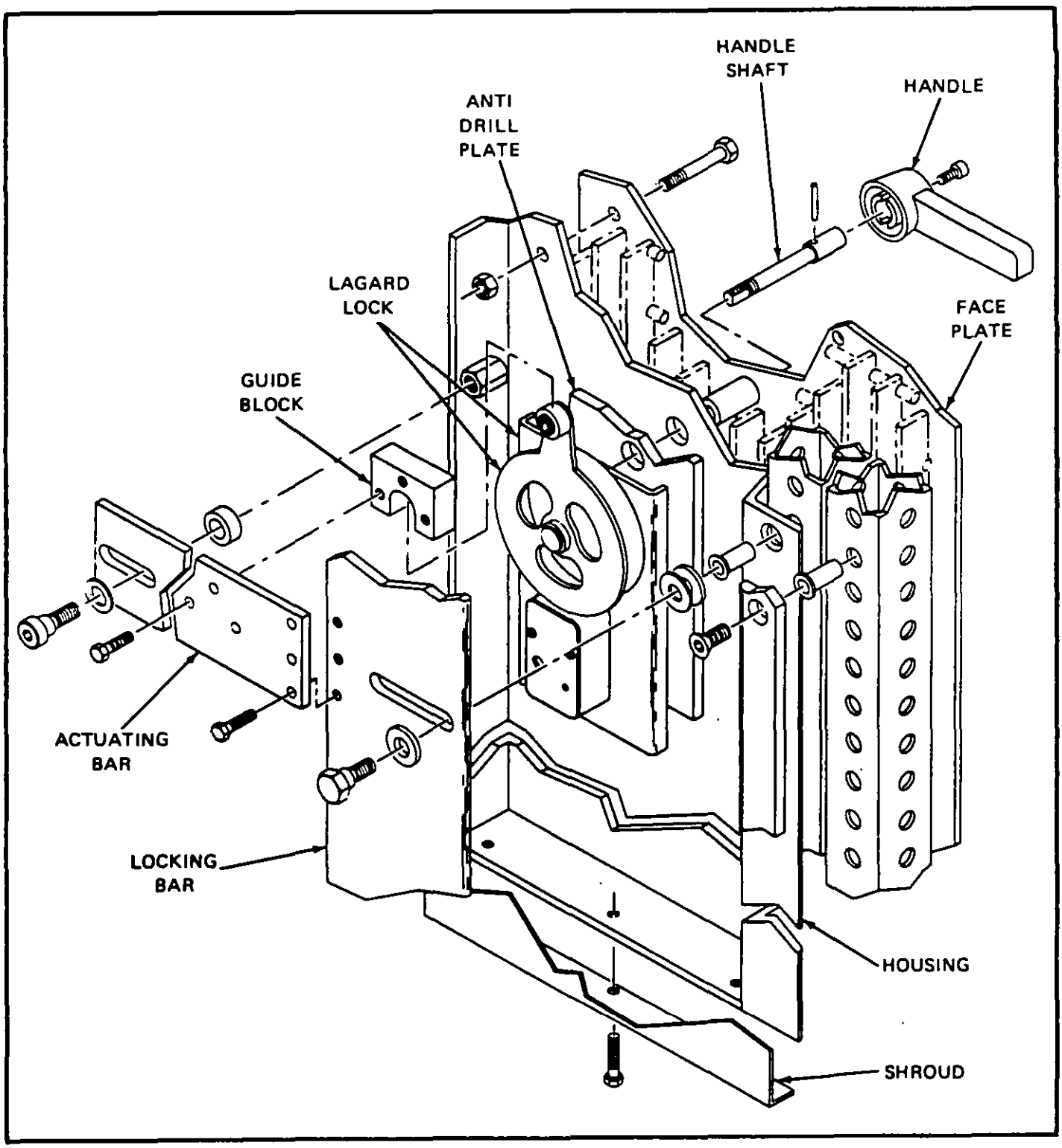


Figure 13. Internal Locking System Assembly

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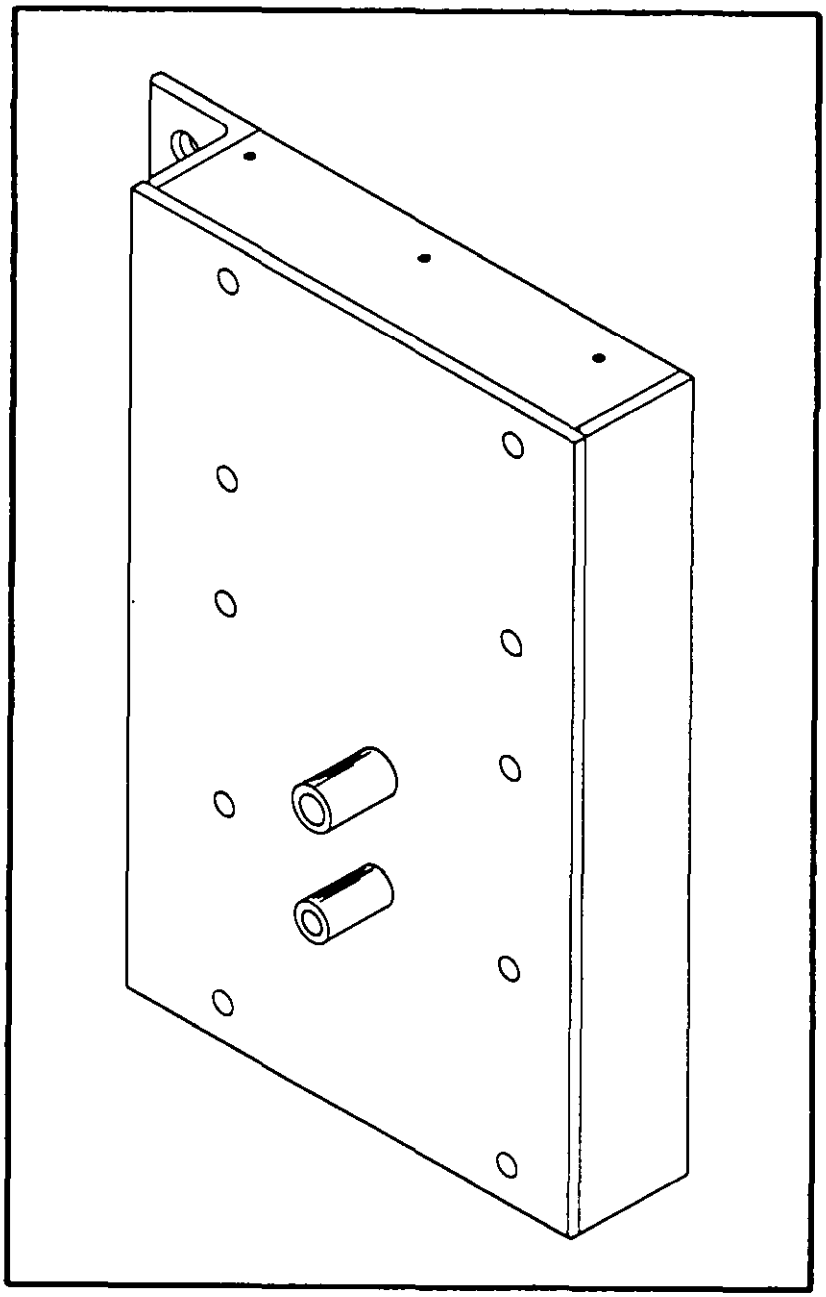


Figure 14. Internal Lock Housing

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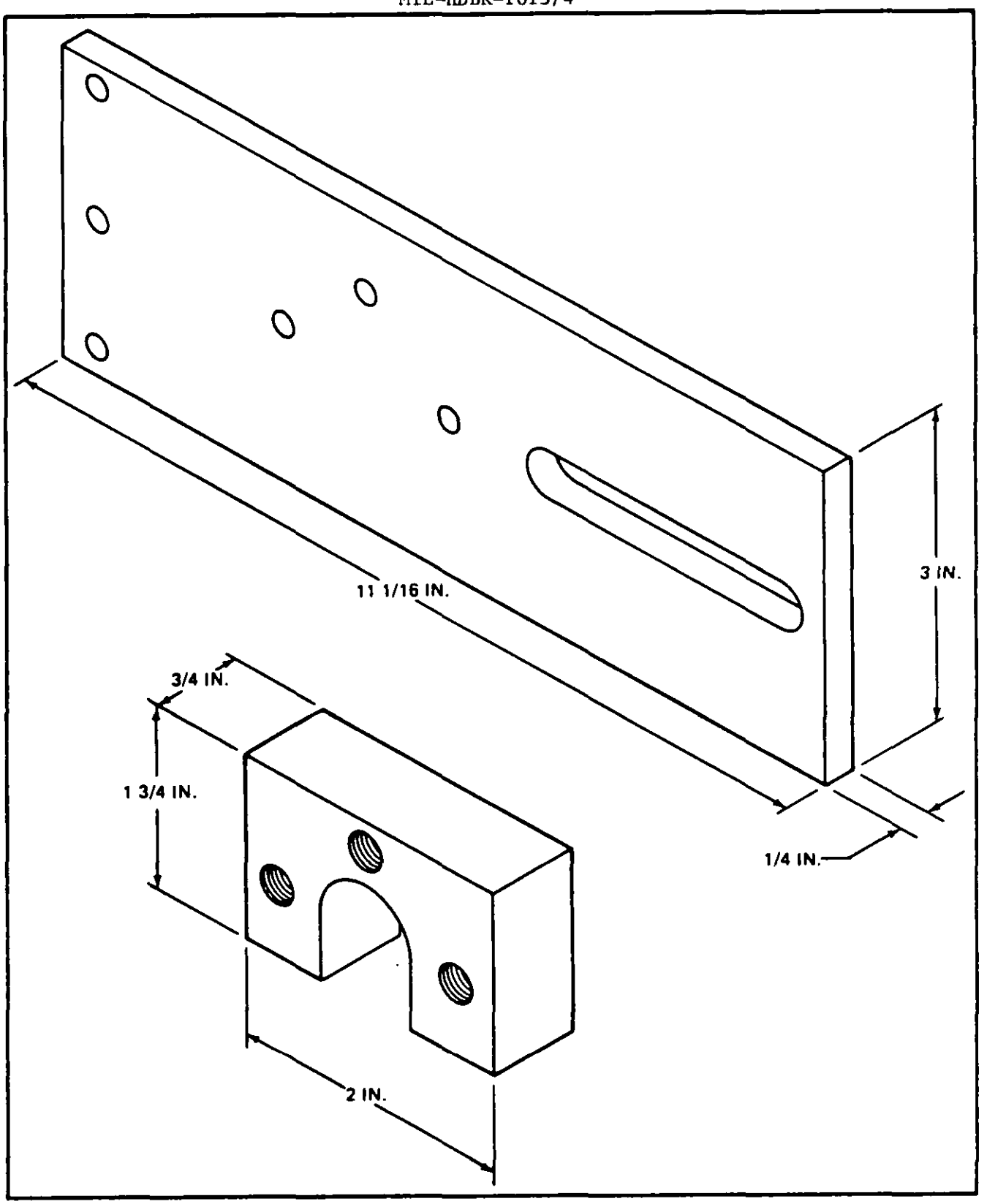


Figure 15. Internal Lock Guideblock and Actuating Bar

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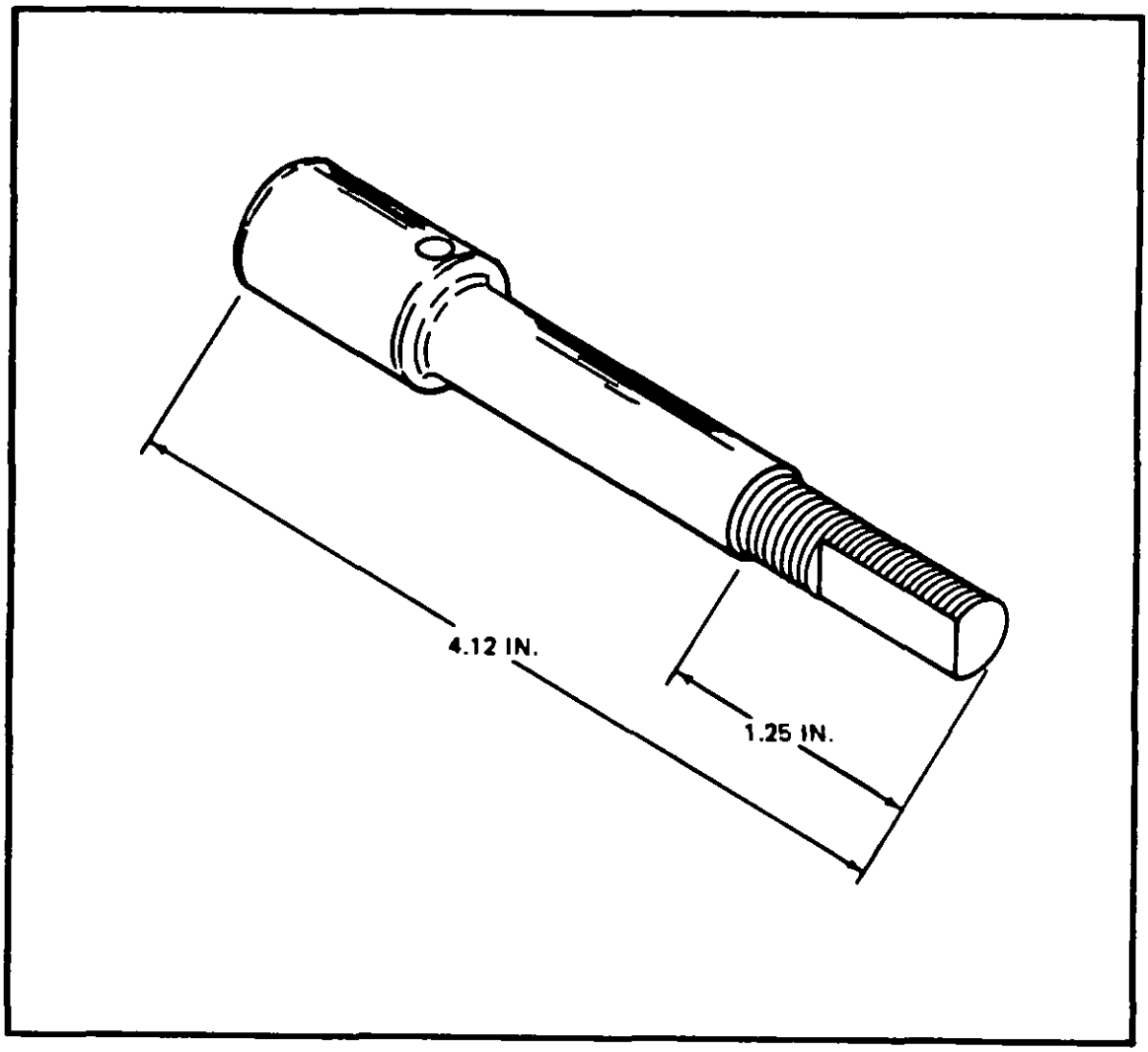


Figure 16. Internal Lock Shaft Handle

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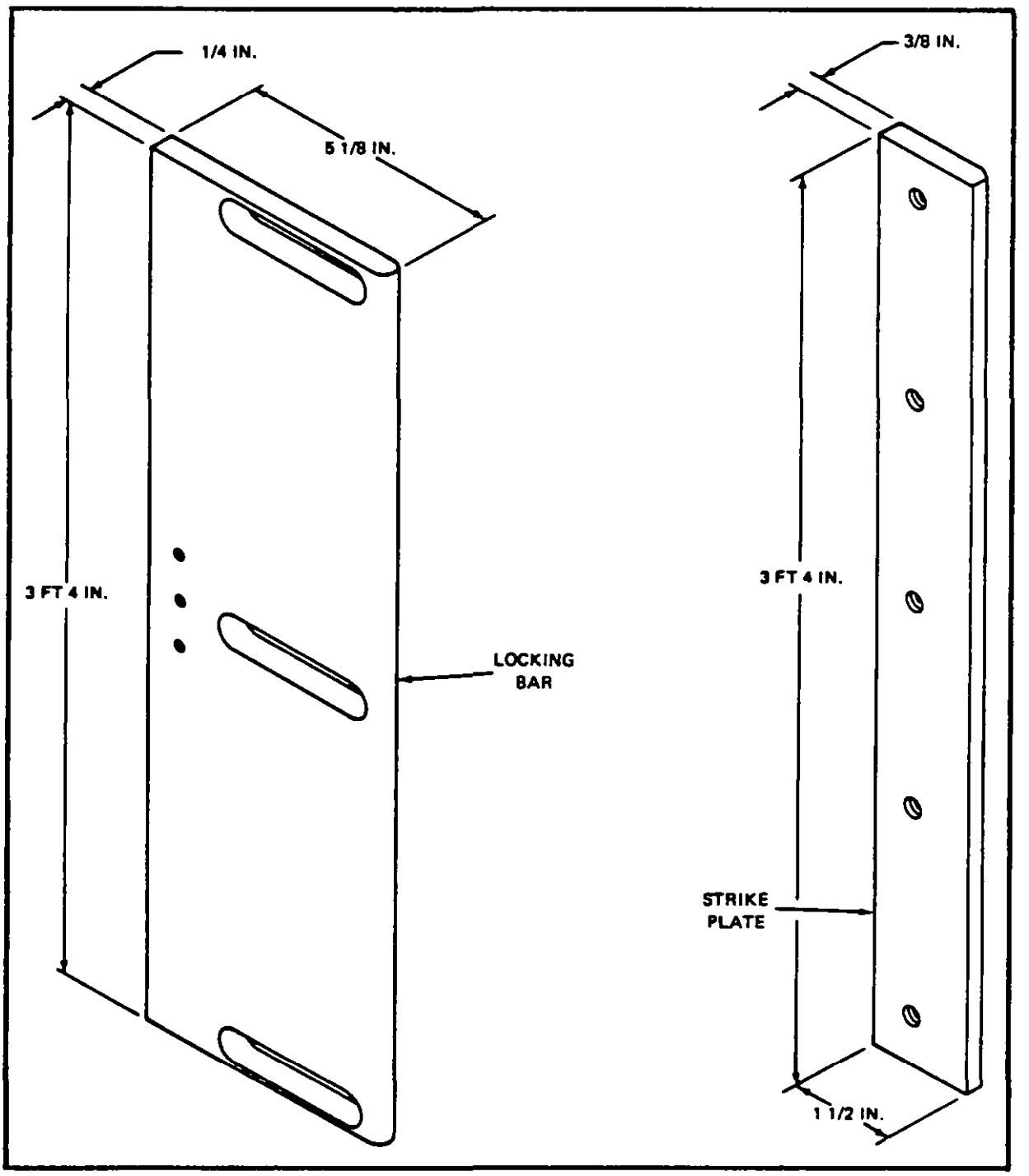


Figure 17. Internal Lock Locking Bar and Strike Plate

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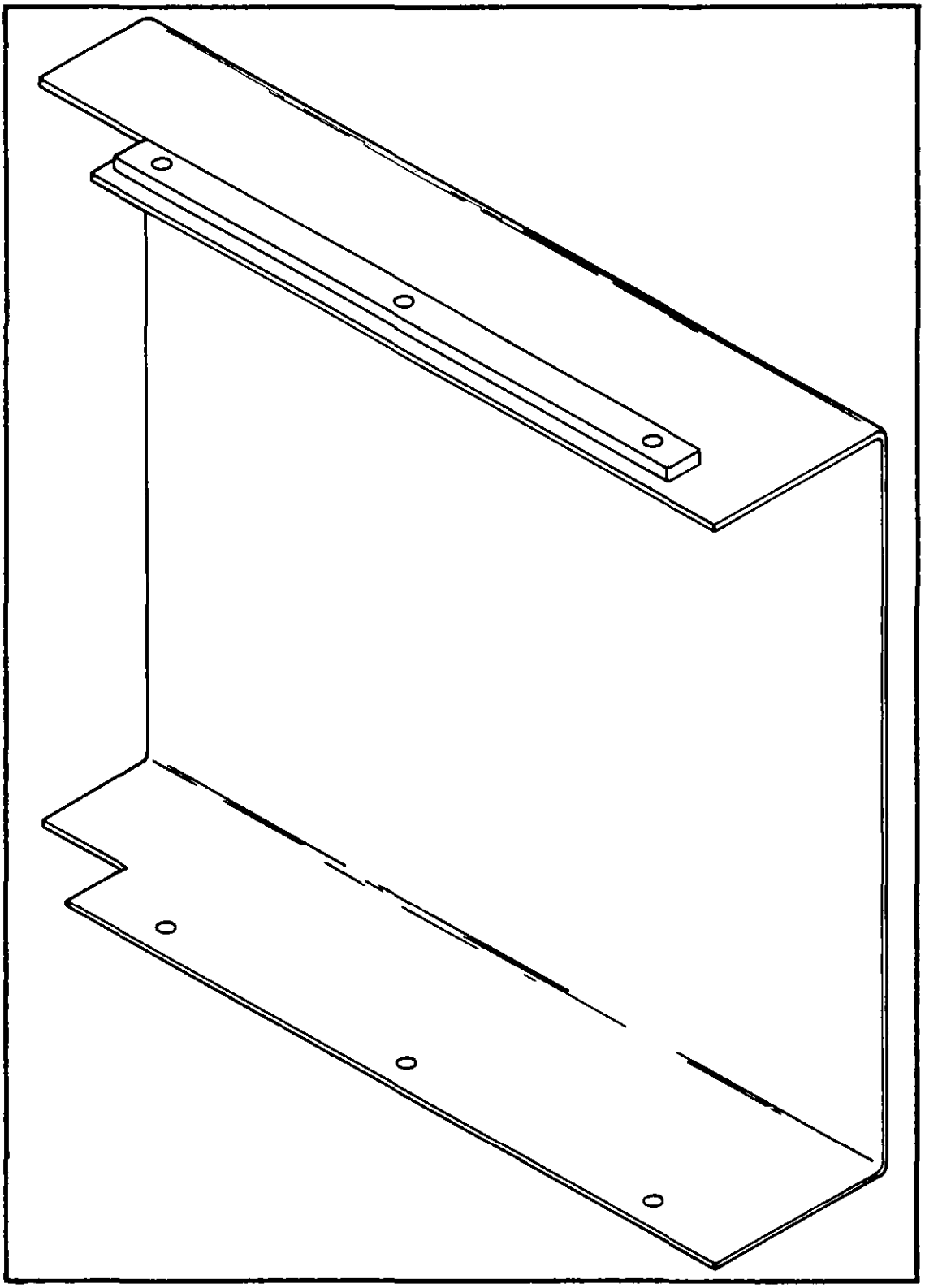


Figure 18. Internal Lock Shroud

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Table 7
Equipment Requirements for Internal Locking System Fabrication

Quantity	Description
1	140-amp AC/4,000-watt welder/power plant
1	15-inch drill press, bench
1	Floor shear, heavy duty
1	8-inch metal lathe
1	Hammer, welders
1	Combination bender, former, and shear
1	4-inch wrench, open end, adjustable
1	Wrench set, 3/8-inch drive with 1/4-, 3/8-, and 5/16-inch sockets
1	5/16-inch hex key wrench
1	Drills, general purpose jobbers in following sizes: 1/4, 9/32 11/32, and 7/16 inch
1	11/16-inch pilot counterbore
1	Drill, size "Z", general purpose jobbers
1	82° 3/8-inch countersink, standard six flute
1	1/4-inch 20 standard tap
1	41/64-inch (0.6406) drill, general purpose jobbers
1	49/64-inch (0.7656) driller, taper length
1	90° 3/4-inch countersink, standard six flute
1	Drilling machine, bench top
4	3-inch "C" clamps

(c) Beam Support: A 4-inch support angle cut from 5- by 5- by 5/16-inch angle reinforced by 1/4-inch bar as detailed in NAVFAC drawing 6155277. The support angle provides support to the spanner beam when the beam is installed. Two support angles are required for each spanner beam.

(d) Hinge Protection Bar: A 2-1/4- by 1/4-inch by 1-foot 2-inch bar bolted to the inside of the door panel behind each installed hinge.

(e) Splice Bar: A 4-inch by 9-foot 8-inch bar used when bolting two panels together to form a door. One bar is required for each panel to panel joint used in fabricating the door.

(f) Spanner Beam: A 19-foot 11-1/2-inch beam cut from a W8X15 beam used to support the roof panels when the length of the enclosure exceeds 10 feet. The 19-foot 11-1/2-inch requirement is for an enclosure of a maximum 20-foot width. For enclosures less than 20 feet wide, the beam length is reduced in 2-foot increments. One beam is required for each additional 10 feet in length of an enclosure. For example, if a 20-foot-long enclosure is installed, one beam would be required in the center, across the width of the enclosure.

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(g) Support Angle: A 9-foot 7-3/4-inch or a 19-foot 7-3/4-inch angle cut from a 2- by 2- by 3/16-inch angle. The support angles are used to attach the roof panels to wall panels or structure walls.

(h) Door Support Angle: A 2-inch support angle is cut from 6- by 3-1/2- by 5/16-inch angle. The support angle, as detailed in NAVFAC drawings 6155276 and 6155278, is required on the inside of the panel that supports the door hinge joint. The support is used to attach the corner of the hinged panel to the floor of the structure.

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Section 5: ASSEMBLY AND INSTALLATION

5.1 Introduction. This section describes how the secure enclosures are assembled and installed within a structure. As described in Section 3, the enclosures can be installed in a variety of configurations. The methods of assembling the panels to form the walls, roofs, and doors of secure enclosures, and methods of attaching enclosures to structure walls and floors are identical regardless of the enclosure configuration selected.

5.2 Resources Required for Installation. Two persons are required for installing the secure enclosure. One person shall be a journeyman mechanic or steelworker and one shall be a qualified forklift (2,000 pounds minimum) operator. Tools and equipment required for installation are listed in Table 8. Commercially available hardware required for assembly and installation is listed in Table 9.

Table 8
Equipment Requirements for Assembly and Installation of Secure Enclosures

Quantity	Description
1 (preferred)	3/8-inch drive impact tool with 1/4-, 3/8-, 5/16-, 9/16-, and 5/8-inch socket wrenches
1 (alternate)	Wrench set 3/8-inch drive, with 1/4-, 3/8-, 5/16-, 9/16-, and 5/8-inch socket wrenches
1	1/2-inch electric hand drill
1 each	1-1/8- (1/2-inch shank) and 9/16-inch masonry drills
4 each	6-, 8-, and 12-inch C-clamps, malleable iron
1	Forklift (2,000 pounds minimum) with crane hooks
1	140-amp AC/4,000-watt welder/power plant

5.3 Enclosure and Locking System Installation.

5.3.1 Enclosure. At the installation location, position all fabricated components, hardware, and other required tools and hardware identified in Tables 8 and 9. Referring to Naval Facilities Engineering Command (NAVFAC) drawings 6155276 and 6155278 for assembly and installation details, proceed as follows:

(a) Lay out secure enclosure installation area with chalk lines. If existing walls are to be used, mark layout on appropriate walls.

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Table 9
Hardware Requirements (Commercial) for
Assembly of Secure Enclosure

Item	Application	Size	Quantity	Source
90° attachment bolt	Roof panel to wall panel	5/16-18	25 per support angle	Unistrut Div, GTE Products Corp. (415) 592-9440 P/N TL061
	Roof panel to wall (existing)		20 per 20-foot support angle	
	Roof panel to wall (existing)		10 per 10-foot support angle	
	Panel to panel, corner connection		35 per corner angle	
	Roof panel to spanner beam		60 per beam (3 per panel)	
	Door panel to door panel		14 per splice bar	
	Panel to clip angle (at floor)		2 per panel	
	Panel to door surface bolt (floor and overhead)		2 per surface bolt	
Flat washer	90° attachment bolt	3/4-OD x 0.05-inch stainless steel	2 per bolt	Commercial
Nut (jam)	90o attachment bolt	5/16-18 stainless steel	2 per bolt	Commercial
Bolt	Panel to panel	3/8 x 16 x 4-1/2-inch	7 per panel	Commercial
Nut	Panel bolt	3/8 x 16	1 per bolt	Commercial
Hex head bolt	Corner angle to wall	3/8-16 x 3/4-inch	12 per corner angle	Commercial
Nondrilling anchor	Anchors	3/8		Commercial
Washer	Panel to floor	1-inch	As required	Commercial

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(b) Using panel-to-floor attachment clip angles as templates, mark holes on floor for 3/8-inch nondrilling concrete anchors. Two clip angles are required per panel. Using 9/16-inch masonry drill, drill 9/16- by 1-9/16-inch holes in floor. Insert anchors in holes.

(c) Bolt two clip angles to bottom of each of two panels.

WARNING

Panels must be securely braced in the upright position before bolting. Injury to personnel and damage to panels may occur if panel falls.

(d) Using forklift and crane hooks, upend two panels to form corner of the enclosure. Brace panels in place with 2 by 4 lumber and C-clamps.

(e) Position corner angle between two panels and fasten panels together using 90° attachment bolts (Figure 19).

(f) Using 3/8- by 3/4-inch bolts with 1-inch washers, fasten clip angles to concrete floor anchors (Figure 19).

(g) Continue around enclosure, one panel at a time, bolting panels together and to floor. Fasten panels together with 3/8-16 by 4-1/2-inch hex head bolts and nuts except at corners (Figure 19).

(h) For enclosures attached to structure walls, use corner and roof support angles as templates and mark holes for concrete anchors. Using 9/16-inch masonry drill, drill holes and insert 3/8-inch anchors.

(i) For panels that support hinge joint or lock, attach hinge or lock side to floor using door support angle.

(j) Brace panels in place as necessary using 2 by 4 lumber and C-clamps.

(k) Clamp roof support angles at upper edge of each row of panels, and bolt to panels on 8-inch centers using 90° attachment bolts, washers, and nuts (Figure 20). Remove clamps.

(l) Mount spanner beams (if required) every 10 feet across width of enclosure. Use two beam supports for each installed spanner beam (Figure 21).

(m) Using forklift and crane hooks, lift and place roof panel between enclosure wall and spanner beam (if used). Bolt panels to support angles and spanner beam with 90° attachment bolts, washers, and nuts on 8- and 12-inch centers. Bolt roof panels to each other on 19-inch centers with 3/8-16 by 4-1/2-inch bolts (Figure 20).

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(n) Using forklift and crane hooks, lift and place door panels(s) in door opening. Using wedges, shims, and C-clamps, position door(s) with 3/8-inch space between door(s) and floor. Shim door(s) a minimum of 5/16 inch apart or from locking wall panel side. Note that maximum available clearance is dependent on accummulation of door panel width tolerances and number of door panels used. Recommend that this distance not be set greater than 1/2 inch. Using 2 by 4 lumber and C-clamps, securely brace doors in place.

(o) Weld six hinges on hinge mounting plates as shown in Figure 22 and NAVFAC drawing 6155276. Clean area. Touchup paint with shop primer.

(p) Bolt six hinge protection bars to doors in accordance with NAVFAC drawing 6155276.

(q) Mark location for door surface bolt below door when in closed position. Drill 1-1/8- by 1-1/4-inch hole in floor using 1-1/8-inch masonry drill.

5.3.2 Locking Systems.

5.3.2.1 Internal Locking System. Using the fabricated components and other hardware requirements detailed in NAVFAC drawings 6155318 and 6155319 and Figure 13, proceed with lock assembly and installation as follows:

(a) Insert antidrill plate behind the lock as indicated in NAVFAC drawing 6155318 and Figure 13.

(b) Assemble boltworks handle through door faceplate and antidrill plate into boltwork mechanism as indicated in the drawings.

(c) Assemble guideblock to actuating bar.

(d) Place guideblock on boltwork actuator and attach actuating bar to housing with 3/8- by 3/4-inch shoulder bolt as shown in drawing.

(e) Using three bolts, 5/6-18- x 1-1/4-inch and three type S31-406 rivnuts, attach locking bar to housing and doorframe.

(f) Attach locking bar to actuating bar as shown in drawing.

(g) Attach strike plate to inactive leaf with five flat-head hex socket bolts as shown in drawing.

(h) Apply liberal amount of lubricating grease to all sliding surfaces. Place key in keylock and, in accordance with instructions provided with the lock, work boltwork system several times to ensure proper operation of locking system.

(i) If double-doors are installed, operate door surface bolts.

(j) Assemble shroud over lock housing as shown in drawing.

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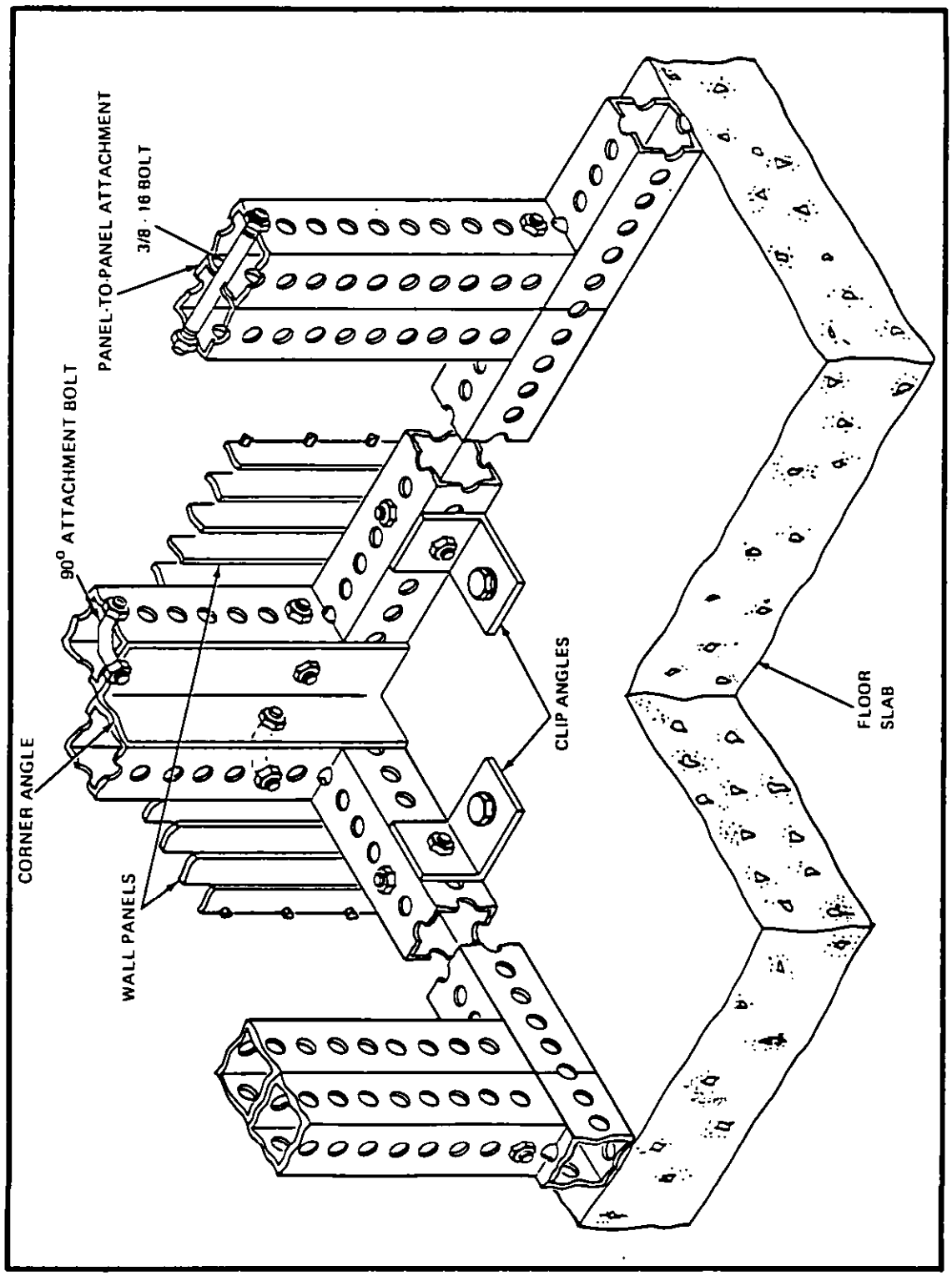


Figure 19. Corner Bolt and Panel-to-Panel Detail

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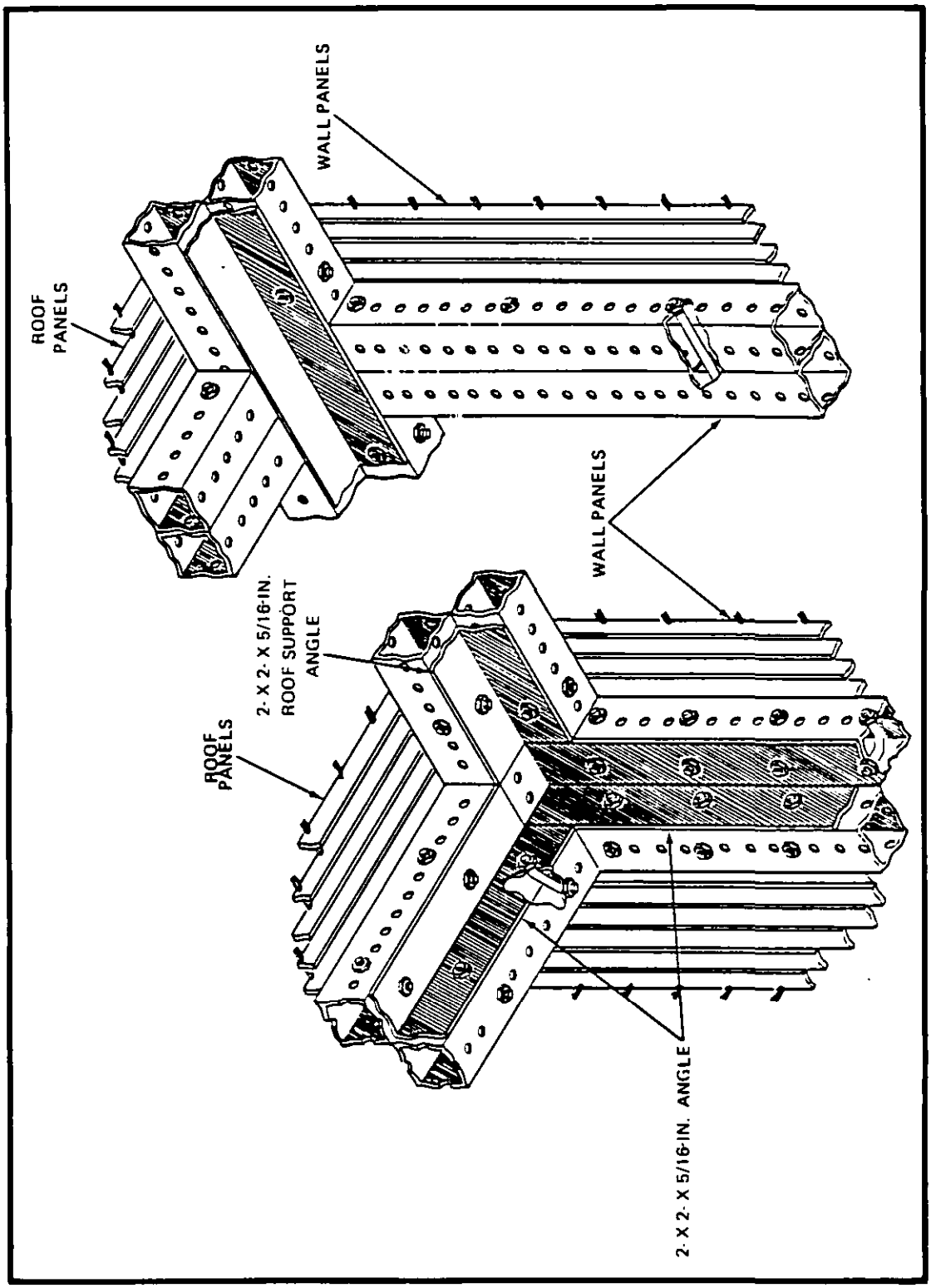


Figure 20. Wall Panel-to-Roof-Panel Joint

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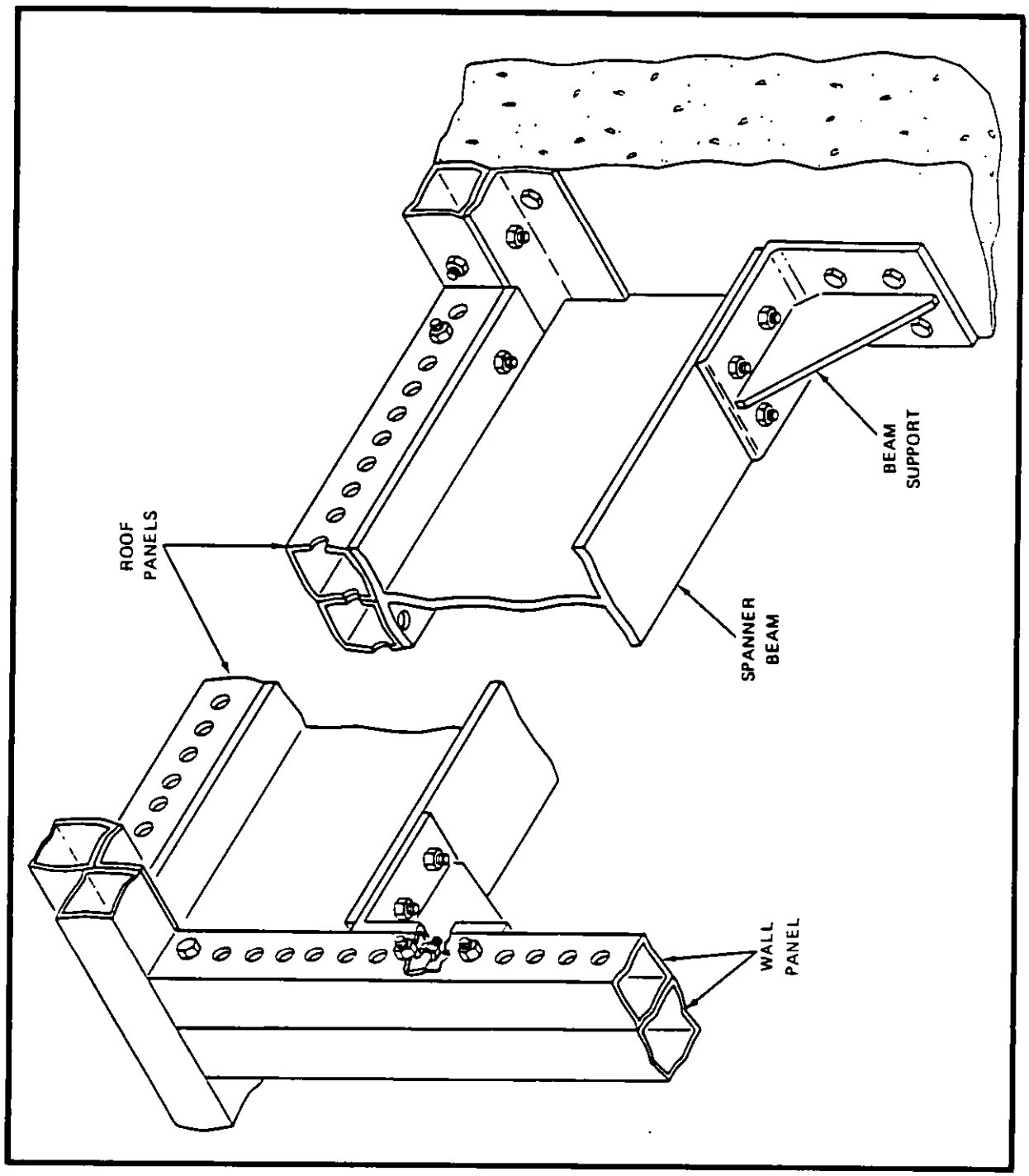


Figure 21. Spanner Beam Installation

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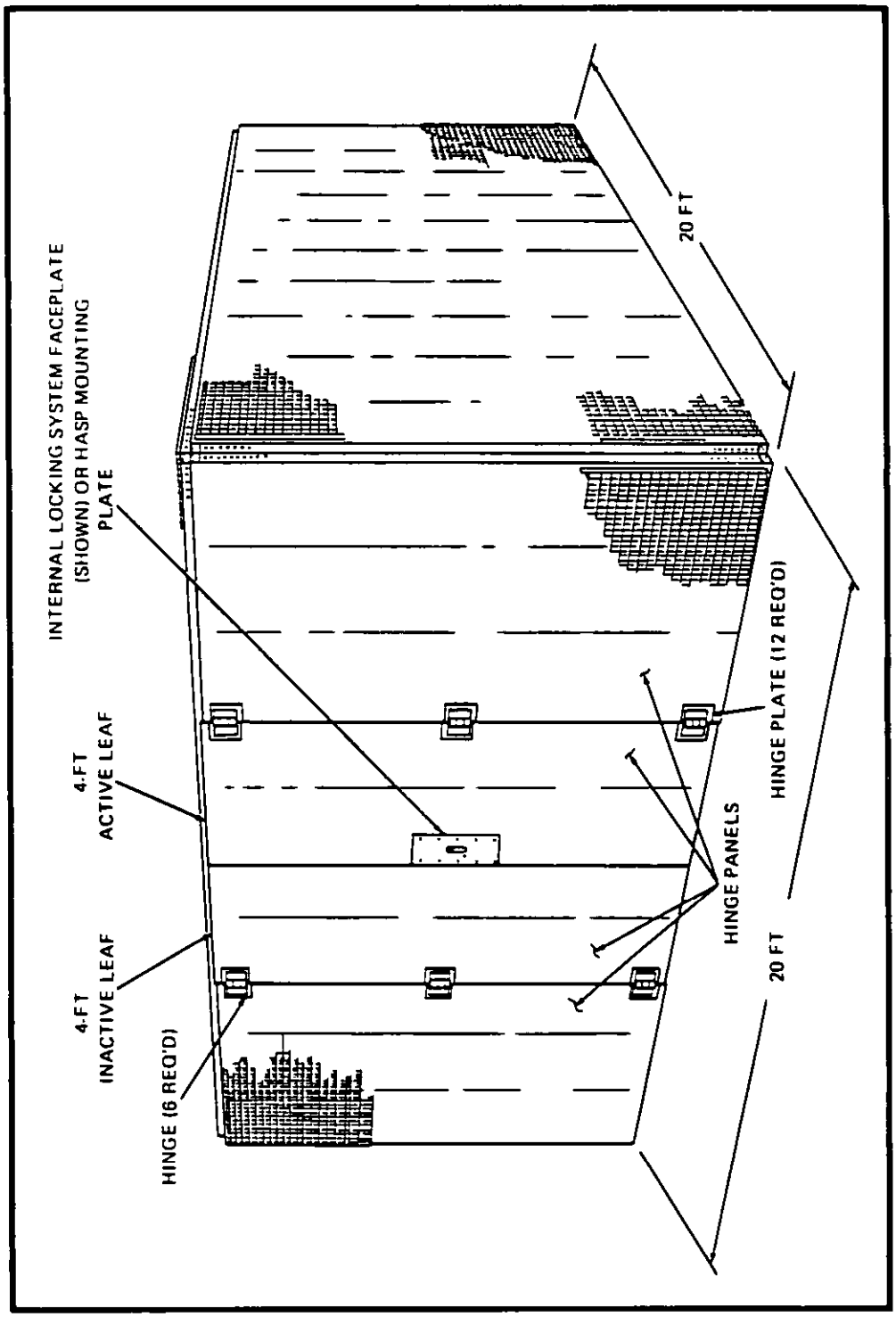


Figure 22. Typical 20- by 20-Foot Enclosure

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5.3.2.2 External Locking System. Using the Naval Ammunition Production Engineering Center (NAPEC) hasp and approved high-security padlock, proceed as follows:

(a) Weld NAPEC hasp to hasp mounting plates on the door(s) and/or wall panel in accordance with NAPEC drawing 0957 or 0958 in locations indicated on NAVFAC drawing 6155278.

(b) Clean and touchup paint with shop primer.

(c) If double-doors are installed, operate door surface bolts.

(d) Install high-security padlock on hasp; check for binding and fit.

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Section 6: SCHEDULED MAINTENANCE

6.1 Introduction. This section provides recommended preventive maintenance procedures to be performed on a scheduled basis for the secure enclosures and associated internal locking system. Maintenance of the high-security padlock, if installed, will be in accordance with separate procedures for the lock and are not repeated herein.

6.2 Procedures. The secure enclosure and internal locking system can be installed in a variety of configurations and environments. Therefore, scheduled maintenance procedures, in particular the recommended intervals, should be modified to suit the installation.

6.2.1 Secure Enclosure Scheduled Maintenance. The enclosure is to be inspected for integrity of assembly, and the doors and hinges lubricated annually. Equipped with 9/16- and 5/8-inch box wrenches, general purpose lubricating oil, and wiping rags, an individual should proceed as indicated in Table 10.

Table 10
Secure Enclosure Scheduled Maintenance

1. Inspect enclosure
   a. Unlock and open door.
      (1) For door with internal locking system:
         (a) Insert key in keyway and turn counterclockwise to unlock boltwork mechanism.
         (b) Turn boltworks operating handle counterclockwise to open boltworks.
         (c) Open door.
      (2) For door with external high-security hasp and padlock:
         (a) Insert key in keyway in bottom of padlock.
         (b) Turn key clockwise to open lock.
         (c) Remove lock from hasp.
         (d) Place lock through hole in bottom hasp and lock.
   b. Visually inspect all bolts and nuts around inside of enclosure. Ensure nuts are attached to bolts and all nuts and bolts are tight. Tighten any found loose. Tighten only enough to ensure snug fit.
2. Lubricate hinges and surface bolts.
   a. Lubricate all door hinges by placing small amount of oil on top of each hinge.
   b. Lubricate two surface bolts located at top and bottom of inactive door leaf by placing small amount of oil on top of each surface bolt. Release bolts and open door.
   c. Open and close doors several times to ensure oil penetrates hinges.
   d. Wipe off any excess oil from hinges and any oil that may have dripped on floor.

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6.2.2 Internal Locking System Scheduled Maintenance. The internal locking system is to be inspected for integrity of assembly, cleaned, and lubricated annually. Tools and materials required to complete maintenance are listed as follows:

- Wrench set, 3/8-inch drive, with 1/4-, 3/8-, and 5/16-inch sockets
- 5/16-inch hex key wrench
- 4-inch adjustable open end wrench
- Brush, cleaning, tool and parts
- Rag, wiping
- Grease, general purpose, wide temperature range MIL-G-81322
- Lubricating powder, molybdenum disulfide, MIL-M-7866, Type II
- Solvent, cleaning trichloroethane
- Drop cloth.

To clean and inspect the internal locking system, an individual equipped with the required tools and materials should proceed as indicated in Table 11.

Table 11
Cleaning and Inspecting Internal Locking System

1. Unlock and open door
   a. Insert key in keyway and turn counterclockwise to unlock boltwork mechanism.
   b. Turn boltworks operating handle counterclockwise to open boltworks.
   c. Open door.
2. Clean internal locking system mechanism.
   a. Remove six 1/4-inch capscrews from shroud cover and remove shroud.
   b. Place drop cloth beneath door under locking system.

WARNING

Avoid prolonged contact with, or inhalation of, cleaning solvent. Avoid use near heat or open flame, and provide adequate ventilation.

   c. Using solvent sparingly, clean excess grease from boltwork mechanism with brush. Wipe away grease and solvent with rags.
3. Inspect boltworks.
   a. Visually inspect boltworks for loose components and/or tolerances while operating mechanism.
   b. Using socket wrenches and hex key, check tightness of all nuts and bolts. Do not attempt to tighten beyond snug fit.

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Table 11 (Continued)

4. Lubricate locking system. a. Place small amount of molybdenum disulfide powder on lock key, and insert key in keyway. Work key back and forth several times (lock and unlock) to ensure that lubricant is worked into lock. b. Place liberal amount of general purpose lubricant on sliding surfaces of boltwork mechanism. Operate boltworks several times with boltwork operating handle to work lubricant into components. 5. Replace shroud over system and attach with six capscrews. 6. Close and relock enclosure doors.

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Section 7: TROUBLESHOOTING

7.1 Introduction. The purpose of this section is to identify the problems likely to occur with the secure enclosure and the internal locking system, probable causes of these problems, and the recommended remedies. Troubleshooting and other required maintenance actions for the externally mounted high-security padlock are covered in other maintenance documents and are not repeated herein.

7.2 Troubleshooting Secure Enclosure and Internal Locking System.

7.2.1 Secure Enclosure. For troubleshooting of the secure enclosure, refer to Table 12.

Table 12
Secure Enclosure Troubleshooting

Problem	Immediate Action	Probable Cause	Corrective Action
Door cannot be opened	Stop, do not force	Locking system inoperable	Check locking system, determine cause and repair
Door binds	Open slowly to enable entry	Panels bent	Check panels, replace as necessary
		Loose bolts on panel supporting doors	Tighten loose bolts

7.2.2 Internal Locking System. For troubleshooting of the internal locking system, refer to Table 13.

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Table 13
Internal Locking System Troubleshooting

Problem	Immediate Action	Probable Cause	Corrective Action
Key will not turn in lock	Stop, do not force key	Lock frozen	Thaw lock
		Tumblers dirty or corroded	Clean and lubricate lock
			Call locksmith
		Foreign matter in lock keyway	Clean lock
			Call locksmith
Boltworks will not not operate	Stop, do not force	Keylock not freeing boltworks	Replace keylock
		Boltworks frozen	Thaw boltworks
Boltworks difficult to operate	Open door	Improper door alignment	Inspect doors and panel attachment

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Section 8: CORRECTIVE MAINTENANCE

8.1 Introduction. This section identifies corrective maintenance actions for the secure enclosure and its internal locking system at the organizational and intermediate levels. Corrective maintenance can be carried out by journeyman mechanics with the tools listed in Table 14. No other special tools or skills are required, with the possible exception of the usual skills and tools of a locksmith for work on the keylock. Corrective maintenance of the high-security padlock, if installed, is described in other maintenance documents and is not repeated herein.

Table 14
Tools and Equipment for Secure Enclosure With Internal Locking System

Quantity	Description
1	4-inch wrench, open end adjustable
1	5/16-inch hex key wrench
1	Wrench set, 3/8-inch drive with 1/4-, 5/16-, 9/16-, and 5/8-inch sockets
1 lb	Grease, wide temperature range conforming to MIL-G-81322
1 tube	Lubricant, molybdenum disulfide technical (powder) lubrication grade MIL-M-7866B

8.2 Adjustments and Alignments.

8.2.1 Secure Enclosure. Adjustment and alignment problems should be corrected by tightening loose nuts or bolts on the enclosure, including floor to wall attachment points and the doors. Should an adjustment or an alignment problem persist, corrective actions will require replacement of bent or otherwise damaged hardware, including the enclosure panels. With the exception of standard hardware such as nuts and bolts, there are no stocked or commercially available parts for the enclosure. Damaged components will require removal and repair or fabrication in a metal working shop.

8.2.2 Internal Locking System. Adjustments and alignments to the internal locking system will require removal of the shroud. Tightening the nuts and bolts used in the assembly of the boltwork mechanism should correct alignment problems. Should a problem persist with no identifiable alignment problems with the enclosure, then one or more of the boltwork components are damaged. The damaged component(s) must be removed and returned to a machine shop for repair or fabrication of a new component. With the exception of the standard, commercially available nuts and bolts, and the LAGARD, Inc., keylock and

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boltwork actuating mechanism, all components of the boltwork mechanism will require machine shop fabrication. After reassembly of the boltworks, the maintenance requirements, included in Section 6, for lubricating the boltworks are to be followed.

8.3 Repair.

8.3.1 Secure Enclosure Repairs to the secure enclosure beyond those identified in Paragraph 8.2.1 should not be required.

8.3.2 Internal Locking System.

8.3.2.1 Key Repair. Corrective maintenance of keys requires replacing damaged or suspect keys as soon as imperfections are detected. Inspection can be encouraged by posting a reminder at the key location. Inspect key at each usage. Ensure that keys are straight, show no cracks or abnormal wear, and show no signs of strain.

8.3.2.2 Lock and Boltwork Actuating Mechanism. Scheduled maintenance in Section 6 requires the locking system to be lubricated annually. Should the lock or boltwork mechanism become difficult to operate at other times, clean and lubricate in accordance with the scheduled maintenance instructions. If there is excessive foreign matter in the keyway, flush with trichloroethane cleaning solvent, and lubricate in accordance with the procedure in Section 6. If these corrective actions do not solve the problem, the services of a professional locksmith will be required.

8.3.2.3 Boltworks. Boltworks repairs beyond those identified in Paragraph 8.2.2 should not be required.

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REFERENCES

Chief of Naval Operations Instruction available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

OPNAVINST 5530.13	Physical Security for Sensitive Conventional AA&E

Military Standards. Department of Defense activities may obtain copies from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MIL-G-81322	Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code Number G-395
MIL-M-7866	Molybdenum Disulfide Technical, Lubrication Grade
MIL-P-29151	Lock and Lock Sets, Exterior, Ordnance, High Security
MIL-P-43607	Padlock, Key Operated, High-Security Shrouded Shackle

Naval Civil Engineering Laboratory (NCEL), Port Hueneme, CA 93043.

DRAWINGS:

6155276	Physical Security: Riveted Steel Grating Cages - Configurations and Assemblies
6155277	Physical Security: Riveted Steel Grating Cages - Panels, Supports, and Surface Bolts
6155278	Physical Security: Welded Steel Grating Cages - Configurations and Assemblies
6155279	Physical Security: Welded Steel Grating Cages - Panels, Supports and Surface Bolts
6155318	Physical Security: Riveted Steel and Welded Steel Grating Cages - Internal Locking System
6155319	Physical Security: Riveted Steel and Welded Steel Grating Cages - Internal Locking System, Face Plates

Naval Facilities Engineering Command Instruction available from Naval Publications and Forms Center, 5801 Tabor Avenue, Phildelphia, PA 19120.

NAVFACINST 5100.11	Command Safety and Health Program

Naval Weapons Support Center, Code 208, Crane, Indiana 47522-5020.

NAPEC Drawing 0957	High-Security Hasp, Hinged, or Sliding Horizontal, RH MK2 MOD 8

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